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Neural network training using ground truth data augmented with map information for autonomous machine applications

Abstract

In various examples, training sensor data generated by one or more sensors of autonomous machines may be localized to high definition (HD) map data to augment and/or generate ground truth data—e.g., automatically, in embodiments. The ground truth data may be associated with the training sensor data for training one or more deep neural networks (DNNs) to compute outputs corresponding to autonomous machine operations—such as object or feature detection, road feature detection and classification, wait condition identification and classification, etc. As a result, the HD map data may be leveraged during training such that the DNNs—in deployment—may aid autonomous machines in navigating environments safely without relying on HD map data to do so.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Provisional Application No. 62/833,185, filed on Apr. 12, 2019, which is hereby incorporated by reference in its entirety. (2) This application is related to U.S. Non-Provisional application Ser. No. 16/814,351, filed on Mar. 10, 2020, U.S. Non-Provisional application Ser. No. 16/514,230, filed on Jul. 17, 2019, and U.S. Non-Provisional application Ser. No. 16/409,056, filed on May 10, 2019, each of which is hereby incorporated by reference in its entirety.

BACKGROUND
(1) For autonomous vehicles to safely navigate through the environment, the vehicles may rely on high definition (HD) maps corresponding to the area in which the vehicle intends to operate. Due to the detailed, three-dimensional, high precision nature of HD maps, navigating according to the HD map data has proven effective for safe navigation of environments where HD map information is available. However, there are cases where HD map information is not available, accurate, or up to date, and/or the HD map may not be relied on—at least exclusively—to aid in navigating through the environment. In such instances, conventional approaches leverage on-board sensors of the vehicles—such as vision sensors (e.g., cameras, LIDAR, RADAR, etc.)—to detect objects, road features (e.g., lane markings, road edges, etc.), free-space boundaries, wait condition information, intersection structure and pose, and/or the like. For example, deep neural networks (DNNs) may be leveraged to process the sensor data from the on-board sensors and compute outputs corresponding to any of the above operations.
(2) However, to be successful and perform at a level of accuracy required for autonomous operation, the DNNs require large amount of diverse training data and ground truth data (e.g., labels, annotations, etc.) corresponding thereto during training. The ground truth data may include, for example, polylines identifying lane markings or road edges, bounding boxes around objects or features of the environment, and/or other ground truth data types. The process of generating the ground truth data requires a substantial amount of manual effort and is typically a significant component of the cost and development time for DNN-based products—e.g., a single set of ground truth data for a training data instance (e.g., an image) may require upwards of twenty minutes of labeling or annotating effort. In addition, manual labeling may not result in ground truth data that enables the DNNs to perform as accurately as possible—e.g., due to human error during labeling or annotating. As a result, where an HD map is unavailable, DNNs may not perform as accurately as desired for safe autonomous operation and/or may require significant cost and manual effort to do so.
SUMMARY
(3) Embodiments of the present disclosure relate to leveraging map information for generating ground truth data for training neural networks for autonomous machine applications. Systems and methods are disclosed that use localization techniques to determine information corresponding to a high definition (HD) map that corresponds to training data (e.g., images, confidence maps, LIDAR data, RADAR data, etc.) captured using one or more sensors of an autonomous machine. The information from the HD map may then be leveraged to generate training labels, annotations, or other ground truth data corresponding to the training data to train one or more neural networks to perform computations corresponding to autonomous machine operations (e.g., object detection, wait condition analysis, road structure determinations, localization, feature detection, etc.).

(4) In contrast to conventional systems, such as those described above, embodiments of the present disclosure combine HD map data with training of deep neural networks (DNNs) to account for flaws or drawbacks in HD maps as well as conventional training processes for DNNs. For example, the limited coverage of accurate and up to date HD maps may be accounted for with DNNs trained using ground truth data generated using HD maps and corresponding to sensor data captured within regions where the HD maps are accurate and up to date. Moreover, the costly and potentially inaccurate ground truth generation process of conventional DNN training techniques may be remedied using automatic generation of accurate ground truth data from HD map information. As a result, training of DNNs may be substantially less costly in terms of time and manual effort and the resulting DNNs that may be used to aid in various operations of autonomous machines may be more accurate and reliable—especially in locations where HD map information is unavailable or not up to date.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The present systems and methods for leveraging map information for generating ground truth data for training neural networks for autonomous machine applications are described in detail below with reference to the attached drawing figures, wherein:
(2) FIG. **1** include an example data flow diagram for a process of localizing training data to high definition (HD) map data to augment or generate labels for training deep neural networks (DNNs), in accordance with some embodiments of the present disclosure;
(3) FIG. **2**A depicts an example visualization of automatically generated ground truth labels corresponding to features of a road, in accordance with some embodiments of the present disclosure;
(4) FIG. **2**B depicts an example visualization of automatically generated ground truth labels corresponding to intersection detection and classification, in accordance with some embodiments of the present disclosure;
(5) FIG. **2**C depicts an example visualization of automatically generated ground truth labels corresponding to features of an environment including poles, road markings, lane lines, road boundaries, and crosswalks, in accordance with some embodiments of the present disclosure;
(6) FIG. **3** depicts an example histogram corresponding to a divergence from a center or rail of a lane of travel, in accordance with some embodiments of the present disclosure;
(7) FIG. **4** is a flow diagram showing a method for localizing training data to HD map data to augment or generate labels for training DNNs, in accordance with some embodiments of the present disclosure;
(8) FIG. **5**A is an illustration of an example autonomous vehicle, in accordance with some embodiments of the present disclosure;
(9) FIG. **5**B is an example of camera locations and fields of view for the example autonomous vehicle of FIG. **5**A, in accordance with some embodiments of the present disclosure;
(10) FIG. **5**C is a block diagram of an example system architecture for the example autonomous vehicle of FIG. **5**A, in accordance with some embodiments of the present disclosure;
(11) FIG. **5**D is a system diagram for communication between cloud-based server(s) and the example autonomous vehicle of FIG. **5**A, in accordance with some embodiments of the present disclosure; and
(12) FIG. **6** is a block diagram of an example computing device suitable for use in implementing some embodiments of the present disclosure.
DETAILED DESCRIPTION
(13) Systems and methods are disclosed related to leveraging map information for generating

ground truth data for training neural networks for autonomous machine applications. Although the present disclosure may be described with respect to an example autonomous vehicle **500** (alternatively referred to herein as "vehicle **500**," "ego-vehicle **500**," "data collection vehicle **500**," or "dynamic actor **500**," an example of which is described with respect to FIGS. **5**A-**5**D, this is not intended to be limiting. For example, the systems and methods described herein may be used by, without limitation, non-autonomous vehicles, semi-autonomous vehicles (e.g., in one or more adaptive driver assistance systems (ADAS)), robots, warehouse vehicles, off-road vehicles, flying vessels, boats, shuttles, emergency response vehicles, motorcycles, electric or motorized bicycles, aircraft, construction vehicles, underwater craft, drones, and/or other vehicle types. In addition, although the present disclosure may be described with respect to autonomous driving or ADAS systems, this is not intended to be limiting. For example, the systems and methods described herein may be used to generate ground truth data for training deep neural networks (DNNs) for implementation in simulation environments, in robotics (e.g., using map information for indoor environments, outdoor environments, warehouse, etc.), aerial systems, boating systems, and/or other technology areas.

(14) With reference to FIG. **1**, FIG. **1** includes an example data flow diagram for a process **100** of localizing training data to high definition (HD) map data to augment or generate labels for training deep neural networks (DNNs), in accordance with some embodiments of the present disclosure. Although the process **100** may be described with respect to the autonomous vehicle **500** and/or an example computing device **600** (FIG. **6**), this is not intended to be limiting. It should be understood that this and other arrangements described herein are set forth only as examples. Other arrangements and elements (e.g., machines, interfaces, functions, orders, groupings of functions, etc.) may be used in addition to or instead of those shown, and some elements may be omitted altogether. Further, many of the elements described herein are functional entities that may be implemented as discrete or distributed components or in conjunction with other components, and in any suitable combination and location. Various functions described herein as being performed by entities may be carried out by hardware, firmware, and/or software. For instance, various functions may be carried out by a processor executing instructions stored in memory.

(15) The process **100** may include generating and/or receiving sensor data **102** from one or more sensors of data collection vehicles **500** (which may be similar to the vehicle **500**, or may include non-autonomous or semi-autonomous vehicles). The sensor data **102** may be used within the process **100** for localization, correlation, and ground truth generation, as well as for input data for a deep neural network (DNN) **116**. The sensor data **102** may include, without limitation, sensor data **102** from any type of sensors, such as but not limited to those described herein with respect to the vehicle **500** and/or other vehicles or objects—such as robotic devices, VR systems, AR systems, etc., in some examples. For non-limiting example, and with reference to FIGS. **5**A-**5**C, the sensor data **102** may include the data generated by, without limitation, global navigation satellite systems (GNSS) sensor(s) **558** (e.g., global positioning system (GPS) sensor(s), differential GPS (DGPS) sensor(s), etc.), RADAR sensor(s) **560**, ultrasonic sensor(s) **562**, LIDAR sensor(s) **564**, inertial measurement unit (IMU) sensor(s) **566** (e.g., accelerometer(s), gyroscope(s), magnetic compass(es), magnetometer(s), etc.), microphone(s) **596**, stereo camera(s) **568**, wide-view camera(s) **570** (e.g., fisheye cameras), infrared camera(s) **572**, surround camera(s) **574** (e.g., 360 degree cameras), long-range and/or mid-range camera(s) **598**, speed sensor(s) **544** (e.g., for measuring the speed of the data collection vehicle and/or distance traveled), and/or other sensor types.

(16) In some examples, the sensor data **102** may include the sensor data generated by one or more forward-facing sensors, side-view sensors, and/or rear-view sensors. This sensor data **102** may be useful for identifying, detecting, classifying, and/or tracking objects around the data collection vehicle within the environment—e.g., for localization, correlating with an HD map **104**, ground truth generation, and/or the like. In embodiments, any number of sensors may be used to

incorporate multiple fields of view (e.g., the fields of view of the long-range cameras **598**, the forward-facing stereo camera **568**, and/or the forward facing wide-view camera **570** of FIG. 5B) and/or sensory fields (e.g., of a LIDAR sensor **564**, a RADAR sensor **560**, etc.).

(17) In some embodiments, the sensor data **102** may include image data representing an image(s), image data representing a video (e.g., snapshots of video), and/or sensor data representing representations of sensory fields of sensors (e.g., depth maps for LIDAR sensors, a value graph for ultrasonic sensors, etc.). Where the sensor data **102** includes image data, any type of image data format may be used, such as, for example and without limitation, compressed images such as in Joint Photographic Experts Group (JPEG) or Luminance/Chrominance (YUV) formats, compressed images as frames stemming from a compressed video format such as H.264/Advanced Video Coding (AVC) or H.265/High Efficiency Video Coding (HEVC), raw images such as originating from Red Clear Blue (RCCB), Red Clear (RCCC), or other type of imaging sensor, and/or other formats. In addition, in some examples, the sensor data **102** may be used within the process **100** without any pre-processing (e.g., in a raw or captured format), while in other examples, the sensor data **102** may undergo pre-processing (e.g., noise balancing, demosaicing, scaling, cropping, augmentation, white balancing, tone curve adjustment, etc., such as using a sensor data pre-processor (not shown)). As used herein, the sensor data **102** may reference unprocessed sensor data, pre-processed sensor data, or a combination thereof.

(18) In addition, the process **100** may include generating and/or receiving map data from a map—such as the HD map **104** (which may be similar to the HD map **522** of FIG. 5C)—accessible by and/or stored by the data collection vehicle **500**. In some embodiments, the HD map **104** and/or a localizer **106** may be components of an HD map manager. The HD map **104** may include, in some embodiments, precision to a centimeter-level or finer, such that the data collection vehicle **500** may rely on the HD map **104** for precise instructions, planning, and localization. The HD map **104** may represent lanes, road boundaries, road shape, elevation, slope, and/or contour, heading information, wait conditions, static object locations, and/or other information. As such, the process **100** may use the information from the HD map **104**—such as locations and shapes of lanes—to generate ground truth data for training the DNN **116**. Although described as an HD map **104** herein, this is not intended to be limiting, and the map data may be generated from any type of map with greater or less precision than an HD map **104**. For example, and without departing from the scope of the present disclosure, map data generated from navigation or GPS applications may be used in addition to or alternatively from an HD map **104**.

(19) The sensor data **102** and the map data (e.g., from the HD map **104**) may be used by a localizer **106** to localize (e.g., for location and/or orientation) the data collection vehicle **500** with respect to the HD map **104**. For example, sensor data **102** generated by location-based sensors—e.g., GNSS sensors **558**—may be used to determine an approximate location of the data collection vehicle **500** on the HD map **104**. This information may be used to determine a region of the HD map **104** that should be analyzed to accurately localize the data collection vehicle **500** within the HD map **104**. For example, sensor data **102** generated by vision-based sensors—e.g., cameras, RADAR sensors **560**, ultrasonic sensors **562**, LIDAR sensor **564**, and/or other sensors—may be used to identify features and/or objects within the environment that have known and accurate locations within region of the HD map **104** determined using the location-based sensors. Once the vehicle **500** is localized within the HD map **104** a first time, the locations and/or orientations of the vehicle **500** over time may be tracked using this same localization technique—e.g., using vision-based sensors to identify objects and/or features having known locations in the HD map **104**—and/or may be tracked using sensor data **102** generated by ego-motion sensors of the vehicle **500** (e.g., IMU sensor(s) **566**, speed sensor(s) **544**, steering sensor(s) **540** tracking steering wheel angles, etc.). For example, once the vehicle **500** is localized initially, the ego-motion sensors may be used to accurately track a change in location and/or orientation of the vehicle **500** over time. However, even where ego-motion sensors are used, the localization techniques using vision-based sensors

may be executed periodically (e.g., every 3 seconds, every 10 seconds, etc.) to re-calibrate the system for localizing the vehicle **500** within the HD map **104**.

(20) In some non-limiting embodiments, the sensor data **102** and/or the information from the HD map **104** may be applied to a coordinate transformer to transform the sensor data **102** and/or the HD map **104** to a coordinate system of the vehicle **500** and/or to transform the map data from the HD map **104** to a coordinate space of the sensor data **102** (e.g., to transform the 3D world-space map data to 2D image- or sensor-space). In order for the coordinate transformer **108** to perform the transformations or shifts described herein, intrinsic (e.g., optical center, focal length, etc.) and/or extrinsic (e.g., location of sensor on vehicle **500**, rotation, translation, etc.) parameters of the sensors may be used to determine a correlation between 2D pixel locations (or other image- or sensor-space locations) and 2D or 3D world-space locations on the HD map **104**.

(21) For example, the coordinate transformer **108** may orient the HD map **104** with respect to the vehicle **500** and/or with respect to a field of view of a sensor that captured the instance of the sensor data **102**. In some embodiments, the coordinate transformer **108** may shift the perspective of the map data with respect to a location and/or orientation of the data collection vehicle **500** and/or a sensor thereof. As such, the portion of the HD map **104** that may be used by the ground truth generator **112** to generate ground truth data may be shifted relative to the vehicle **500** (e.g., with the data collection vehicle **500** at the center, at (x, y) coordinates of (0, 0), where y is a longitudinal dimension extending from front to rear of the vehicle and x is a lateral dimension perpendicular to y and extending from left to right of the vehicle) and/or a sensor thereof (e.g., to a field of view or sensory field of the sensor that generated the instance of the sensor data **102** corresponding to the HD map **104**). In some embodiments, in addition to or alternatively from shifting the perspective or coordinate system with respect to the vehicle **500** and/or a sensor thereof, the coordinate transformer **108** may shift the perspective to a same field of view for each type of data. For example, where the HD map **104** may generate data from a top-down perspective of the environment, the sensors that generate the sensor data **102** may do so from different perspectives—such as front-facing, side-facing, angled downward, angled upward, etc. As such, to generate ground truth data from a same perspective, the coordinate transformer **108** may adjust the sensor data **102** and/or the map data to a same perspective. In some non-limiting embodiments, each of the sensor data **102** and the HD map **104** may be shifted to a top-down view perspective or coordinate system of the HD map **104**, a coordinate system or perspective of the sensor data **102**, and/or another perspective or coordinate system.

(22) In addition to or alternatively from the coordinate transformer **108** shifting or transforming the coordinate system of the HD map **104** to that of the vehicle **500** and/or a sensor thereof, the coordinate transformer **108** may, in some embodiments, shift or transform the map data to a coordinate system or dimension of the sensor data **102**. For example, where the DNN **116** is trained to compute outputs **118** in 2D image-space, the map data may be transformed or shifted from 2D or 3D world-space coordinates to 2D image- or sensor-space coordinates. As another example, where the DNN **116** is trained to compute outputs **118** in 3D world-space, the map data may not be transformed or shifted to 2D image-space, even where the sensor data **102** input to the DNN **116** represents sensor data representations in 2D space. As such, the DNN **116** may be trained to compute the outputs **118** in 2D or 3D world-space coordinates and, as a result, the ground truth generated for training the DNN **116** may correspond to a 2D or 3D world-space coordinate system. In at least some embodiments, some (e.g., features, objects, etc., represented thereby) or all of the sensor data **102** that the ground truth corresponds to may be converted from 2D image- or sensor-space to 2D or 3D world-space by the coordinate transformer **108** in order to generate ground truth that corresponds to the 2D or 3D world-space.

(23) Once localization has been performed by the localizer **106**, and/or the sensor data **102** and/or the map data have been transformed or shifted by the coordinate transformer **108**, a correlator **110** (which may include or alternatively be referred to herein as a feature determiner **110**) may correlate

the map data with the sensor data **102**. For example, depending on the type of ground truth data to be generated, the correlator may determine correlations between features and/or objects as represented by the map data and the features and/or the objects as represented by the sensor data **102**. As an example, where the DNN **116** is trained to predict locations of lane lines, dividers, and/or other features of the driving surface, the map data representing the lane lines, dividers, and/or other features may be correlated with the sensor data **102** at each sensor data instance (e.g., at each image or frame) (e.g., similar to visualization **200** of FIG. **2**A). As another example, where the DNN **116** is trained to generate trajectories, as described herein, the correlator **110** may determine rails (e.g., centers) of lanes corresponding to ground truth trajectories such that the ground truth generator **112** may generate the final ground truth trajectories that more closely correspond to or are centered on the rails of the lanes (e.g., similar to the visualization **200** of FIG. **2**A). As a further example, where the DNN **116** is trained to generate outputs corresponding to intersection (e.g., bounding shape vertices corresponding to a bounding shape encompassing an intersection), the correlator **110** may determine each of the features of the intersection (e.g., traffic lights, traffic signs, labels or markings on the driving surface, etc.) that correspond to an intersection such that the ground truth generator **112** may generate bounding shapes that encompass each of the features (e.g., similar to visualization **240** of FIG. **2**B).
(24) The ground truth generator **112** may generate ground truth data using the sensor data **102** and/or the map data from the HD map **104**, according to the outputs **118** the DNN **116** is trained to compute and the format of the outputs **118**. For example, where the DNN **116** is trained to generate outputs corresponding to points of a polyline defining a lane, the ground truth generator **112** may generate the ground truth data from the map data by determining points (at some increment) along the lane markings from the map data, and associating those points with points of a polyline. As such, when the DNN **116** generates the outputs **118** that correspond to the point of the polyline, the ground truth data generated by the ground truth generator **112** may be compared by a training engine **114**—e.g., using one or more loss functions—to the points from the outputs **118** in order to determine updates to parameters (e.g., weights and biases) of the DNN **116**.
(25) Similarly, where the DNN **116** is trained to predict outputs **118** corresponding to bounding shapes of intersections, the ground truth generator **112** may generate the ground truth bounding shapes according to the format of the predictions of the DNN **116**. For example, the DNN **116** may be trained to output the bounding shape coordinates as pixel locations of two or more vertices of the bounding shape in 2D image-space and, as a result, the ground truth generator **112** may generate the ground truth data according to this format—e.g., by generating ground truth data corresponding to the locations of two or more vertices of the bounding shapes. As another example, the DNN **116** may be trained to output the bounding shape coordinates as pixel locations of a centroid of the bounding shape and dimensions (e.g., height and width) in 2D image-space and, as a result, the ground truth generator **112** may generate the ground truth data according to this format—e.g., by generating ground truth data corresponding to the location of a centroid and dimensions of the bounding shapes. As a further example, the DNN **116** may be trained to output the bounding shape coordinates as locations of two or more vertices of the bounding shape in 3D world-space and, as a result, the ground truth generator **112** may generate the ground truth data according to this format—e.g., by generating ground truth data corresponding to the locations of two or more vertices of the bounding shapes in 3D world-space.
(26) In some embodiments, depending on any pre-processing of the sensor data **102**, the ground truth generator **112** may compensate for the pre-processing in the generating of the ground truth data. For example, where instances of sensor data **102** (e.g., images, depth maps, etc.) are pre-processed—e.g., by adjusting spatial resolutions, flipping, rotating, cropping, zooming, augmenting, etc.—the ground truth generator **112** may account for these changes in generating the ground truth data. For example, in some embodiments, each instance of the sensor data **102** may be adjusted in some way. In such examples, the ground truth generator **112** may adjust the ground

truth data accordingly for each instance of the ground truth data (e.g., where each image represented by the sensor data **102** is cropped, each instance of the ground truth data may be adjusted to account for the cropping). In other examples, some instances of the sensor data **102** may be adjusted while others may not—e.g., to train the DNN **116** not to over-fit and to accurately compute the outputs **118** across any sensor data instance variations. In such examples, the ground truth generator **112** may receive data representative of the adjustments to instances of the sensor data **102**, and may compensate for the adjustments when generating the ground truth data. For example, if a first instance of sensor data **102** is unchanged, the ground truth data may be generated normally, but where a second instance of the sensor data **102** is rotated prior to input to the DNN **116**, the ground truth generator **112** may account for this by similarly rotating the ground truth data to correspond to the rotated instance of the sensor data **102**.
(27) Once the ground truth data is generated—automatically, in embodiments—the ground truth data may be used by the training engine **114** to train the DNN **116**. For example, the sensor data **102** may be applied to the DNN **116**, the DNN **116** may generate the outputs **118**, the training engine **114** may analyze the outputs **118** in view of the ground truth data from the ground truth generator **112** using one or more loss functions, and the computations of the training engine **114** may be used to update the DNN **116** until the DNN **116** converges to a desirable or acceptable accuracy.
(28) The DNN **116** may include any type of DNN or machine learning model, depending on the embodiment. For example, and without limitation, the DNN **116** may include any type of machine learning model, such as a machine learning model(s) using linear regression, logistic regression, decision trees, support vector machines (SVM), Naïve Bayes, k-nearest neighbor (Knn), K means clustering, random forest, dimensionality reduction algorithms, gradient boosting algorithms, neural networks (e.g., auto-encoders, convolutional, recurrent, perceptrons, long/short term memory/LSTM, Hopfield, Boltzmann, deep belief, deconvolutional, generative adversarial, liquid state machine, etc.), lane detection algorithms, computer vision algorithms, and/or other types of machine learning models.
(29) As an example, such as where the DNN **116** includes a CNN, the DNN **116** may include any number of layers. One or more of the layers may include an input layer. The input layer may hold values associated with the sensor data **102** (e.g., before or after post-processing). For example, when the sensor data **102** is an image, the input layer may hold values representative of the raw pixel values of the image(s) as a volume (e.g., a width, a height, and color channels (e.g., RGB), such as 32×32×3).
(30) One or more layers may include convolutional layers. The convolutional layers may compute the output of neurons that are connected to local regions in an input layer, each neuron computing a dot product between their weights and a small region they are connected to in the input volume. A result of the convolutional layers may be another volume, with one of the dimensions based on the number of filters applied (e.g., the width, the height, and the number of filters, such as 32×32×12, if 12 were the number of filters).
(31) One or more of the layers may include a rectified linear unit (ReLU) layer. The ReLU layer(s) may apply an elementwise activation function, such as the max (0, x), thresholding at zero, for example. The resulting volume of a ReLU layer may be the same as the volume of the input of the ReLU layer.
(32) One or more of the layers may include a pooling layer. The pooling layer may perform a down sampling operation along the spatial dimensions (e.g., the height and the width), which may result in a smaller volume than the input of the pooling layer (e.g., 16×16×12 from the 32×32×12 input volume).
(33) One or more of the layers may include one or more fully connected layer(s). Each neuron in the fully connected layer(s) may be connected to each of the neurons in the previous volume. The fully connected layer may compute class scores, and the resulting volume may be 1×1×number of

classes. In some examples, the CNN may include a fully connected layer(s) such that the output of one or more of the layers of the CNN may be provided as input to a fully connected layer(s) of the CNN. In some examples, one or more convolutional streams may be implemented by the DNN **116**, and some or all of the convolutional streams may include a respective fully connected layer(s).
(34) In some non-limiting embodiments, the DNN **116** may include a series of convolutional and max pooling layers to facilitate image feature extraction, followed by multi-scale dilated convolutional and up-sampling layers to facilitate global context feature extraction.
(35) Although input layers, convolutional layers, pooling layers, ReLU layers, and fully connected layers are discussed herein with respect to the DNN **116**, this is not intended to be limiting. For example, additional or alternative layers may be used in the DNN **116**, such as normalization layers, SoftMax layers, and/or other layer types.
(36) In embodiments where the DNN **116** includes a CNN, different orders and numbers of the layers of the CNN may be used depending on the embodiment. In other words, the order and number of layers of the DNN **116** is not limited to any one architecture.
(37) In addition, some of the layers may include parameters (e.g., weights and/or biases), such as the convolutional layers and the fully connected layers, while others may not, such as the ReLU layers and pooling layers. In some examples, the parameters may be learned by the DNN **116** during training. Further, some of the layers may include additional hyper-parameters (e.g., learning rate, stride, epochs, etc.), such as the convolutional layers, the fully connected layers, and the pooling layers, while other layers may not, such as the ReLU layers. The parameters and hyper-parameters are not to be limited and may differ depending on the embodiment.
(38) Now referring to FIGS. **2**A-**2**C, FIGS. **2**A-**2**C include example visualizations representing various instances of automatic ground truth generation with the process **100**. The visualizations described herein are for example purposes only, and are not intended to limit the scope of the present disclosure. With reference to FIG. **2**A, FIG. **2**A depicts an example visualization **200** of automatically generated ground truth labels corresponding to features of a road, in accordance with some embodiments of the present disclosure. The visualization **200** may represent an instance of the sensor data **102** (e.g., an image) and the corresponding ground truth data that may be generated using the map data from the HD map **104** and/or trajectory information from ego-motion sensor data. For example, where the DNN **116** is trained to predict features of the road—e.g., rails or centers of lanes, lane dividers, road boundaries, etc.—the ground truth data generated from the map data may include each of the ground truth labels in the visualization **200**. In one or more embodiments, ground truth labels may include all labels except for ego-trajectory **202** (which is described in more detail herein). For example, after the vehicle **500** is localized to the HD map **104**, the coordinate transformer **108** orients the map data with respect to the vehicle **500**, a sensor thereof, and/or a coordinate system or dimensional space of the sensor data **102**, the correlator **110** may determine each road boundary **204** (e.g., **204**A and **204**B), each lane divider **206** (e.g., **206**A-**206**D), each lane rail **210** (e.g., **210**A-**210**C, **208**), and/or other features of the road and their corresponding locations with respect to the instance of the sensor data. The ground truth generator **112** may then generate the ground truth data from each of the labels corresponding to these features in the format the DNN **116** is trained to predict the outputs, as described herein.
(39) As another example, and again with respect to FIG. **2**A, the map data from the HD map may be used to augment other ground truth data. For example, where the ego-trajectory **202** is automatically generated using ego-motion sensors, the ego-trajectory **202** may be adjusted to more closely correspond to the lane rail **208** corresponding to the lane of travel of the vehicle **500**. As such, the ground truth generator **112** may generate a final ground truth trajectory—or data representative thereof, such as points along a polyline defining the final ground truth trajectory—that corresponds to the lane rail **208** instead of the ego-trajectory **202**. The ego-trajectory may, in some non-limiting embodiments, be generated using methods and systems as described in U.S. Non-Provisional application Ser. No. 16/409,056, filed on May 10, 2019, which is hereby

incorporated by reference in its entirety. For example, and with reference to FIG. **3**, because the ego-trajectory **202** may be generated based on a driven path of a human driver, the ego-trajectory may stray from a center or rail of the lane of travel. Chart **300** of FIG. **3** illustrates an amount of shift corresponding to generated ego-trajectories, depicting trajectories that stray from a center or rail of the lane of travel by upwards of half of a meter or more. Where these ego-trajectories are used directly to train the DNN **116** to compute future trajectories for a vehicle, the compute trajectories in deployment may similarly include shifts from rail or center of the lane of travel. As such, by augmenting the ego-trajectory using the map data to generate ground truth data that more closely follow a center or rail of the lane of travel, the compute trajectories for a vehicle in deployment may also more closely follow centers or rails of lanes of travel. In addition, as a trajectory may include changing lanes, making turns, and/or the like, the adjustments to the ego-trajectories for ground truth generation may also enable the turns, lane changes, and/or other maneuvers of the computed trajectories of the DNN **116** to shift from centers or rails of lanes of travel to centers or rails of other lanes of travel. As a result, the compute trajectories may place the vehicle in deployment a safer distance from surrounding objects as the vehicle more closely traverses the driving surface along centers or rails of lanes of travel.

(40) Now referring to FIG. **2**B, FIG. **2**B depicts an example visualization **240** of automatically generated ground truth labels corresponding to intersection detection and classification, in accordance with some embodiments of the present disclosure. For example, the DNN **116** may be trained to generate outputs **118** representative of bounding shapes (e.g., bounding shape **242** of the visualization **240**) corresponding to intersections, classifications corresponding to the intersections (e.g., intersection, stoplight controlled), distances to intersections, and/or other information corresponding to the intersections. Because the map data from an HD map may represent features of the intersection and locations thereof, the map data may be used to generate the ground truth data. For example, where the bounding shape **242** is to encompass each feature of the intersection —e.g., a traffic light **244**, crosswalks **246**A and **246**B, intersection entry lines **248**, etc.—the locations and presence of these features may be determined from the map data and the correlator **110** may determine the dimensions of the bounding shape **242** that encompasses all of them, a classification(s) for the intersection, and a distance to the intersection (e.g., to an entry line from the intersection entry lines **248**). As such, the ground truth generator **212** may generate the ground truth data corresponding to the classification information, the bounding shape dimensions and/or vertices, and/or distances to the intersection. In some non-limiting embodiments, the ground truth generated for the intersection may be similar to that of the ground truth data generated in U.S. Non-Provisional application Ser. No. 16/814,351, filed on Mar. 10, 2020, which is hereby incorporated by reference in its entirety.

(41) With reference now to FIG. **2**C, FIG. **2**C depicts an example visualization **280** of automatically generated ground truth labels corresponding to features of an environment including poles, road markings, lane lines, road boundaries, and crosswalks, in accordance with some embodiments of the present disclosure. For example, the DNN **116** may be trained to compute the outputs **118** corresponding to features—e.g., lines, markings, poles, etc.—of the environment. As such, the map data from the HD map **104** may be used to generate the ground truth data that may represent poles **282**, crosswalks **284**, lane markings **286**, road markings **288**, and/or other features of the environment. As such, after localization, coordinate transformation(s), and/or correlation, the ground truth generator **112** may generate the ground truth data corresponding to each of the features of the environment that the DNN **116** is trained to compute. In non-limiting examples, the ground truth generated may be similar to that described in U.S. Non-Provisional application Ser. No. 16/514,230, filed on Jul. 17, 2019, which is hereby incorporated by reference in its entirety.

(42) Now referring to FIG. **4**, each block of method **400**, described herein, comprises a computing process that may be performed using any combination of hardware, firmware, and/or software. For instance, various functions may be carried out by a processor executing instructions stored in

memory. The method **400** may also be embodied as computer-usable instructions stored on computer storage media. The method **400** may be provided by a standalone application, a service or hosted service (standalone or in combination with another hosted service), or a plug-in to another product, to name a few. In addition, method **400** is described, by way of example, with respect to the process **100** of FIG. **1** and the autonomous vehicle **500** of FIGS. **5**A-**5**D. However, these methods may additionally or alternatively be executed by any one system or within any one process, or any combination of systems and processes, including, but not limited to, those described herein.
(43) FIG. **4** is a flow diagram showing a method **400** for localizing training data to HD map data to augment or generate labels for training DNNs, in accordance with some embodiments of the present disclosure. The method **400**, at block B**402**, includes receiving HD map data corresponding to a region including a location of a dynamic actor at a time. For example, map data corresponding to a region of the HD map **104** including the data collection vehicle **500** may be generated and/or received—e.g., using an HD map manager.
(44) The method **400**, at block B**404**, includes localizing the dynamic actor with respect to the HD map data. For example, the localizer **106** may localize the data collection vehicle **500** within the HD map **104**.
(45) The method **400**, at block B**406**, includes receiving sensor data generated by a sensor of the dynamic actor at a time. For example, an instance of the sensor data **102** generated by a sensor of the data collection vehicle **500** may be generated and/or received.
(46) The method **400**, at block B**408**, includes generating, based at least in part on the HD map data and the localizing, ground truth data corresponding to the sensor data. For example, the ground truth generator **112** may generate the ground truth data corresponding to the instance of the sensor data **102** using the map data. In some embodiments, this may include transforming or shifting a coordinate system of the HD map **104** to a coordinate system of the vehicle **500**, correlating the map data with the sensor data **102**, and/or other processes described herein with respect to the coordinate transformer **108**, the correlator **110**, and/or the ground truth generator **112**.
(47) The method **400**, at block B**410**, includes training a neural network using the sensor data and the ground truth data. For example, the DNN **116** may be trained—e.g., using the training engine **114**—using the sensor data **102** and the ground truth data generated using the ground truth generator **112**.
(48) Example Autonomous Vehicle
(49) FIG. **5**A is an illustration of an example autonomous vehicle **500**, in accordance with some embodiments of the present disclosure. The autonomous vehicle **500** (alternatively referred to herein as the “vehicle **500**”) may include, without limitation, a passenger vehicle, such as a car, a truck, a bus, a first responder vehicle, a shuttle, an electric or motorized bicycle, a motorcycle, a fire truck, a police vehicle, an ambulance, a boat, a construction vehicle, an underwater craft, a drone, and/or another type of vehicle (e.g., that is unmanned and/or that accommodates one or more passengers). Autonomous vehicles are generally described in terms of automation levels, defined by the National Highway Traffic Safety Administration (NHTSA), a division of the US Department of Transportation, and the Society of Automotive Engineers (SAE) “Taxonomy and Definitions for Terms Related to Driving Automation Systems for On-Road Motor Vehicles” (Standard No. J3016-201806, published on Jun. 15, 2018, Standard No. J3016-201609, published on Sep. 30, 2016, and previous and future versions of this standard). The vehicle **500** may be capable of functionality in accordance with one or more of Level 3-Level 5 of the autonomous driving levels. For example, the vehicle **500** may be capable of conditional automation (Level 3), high automation (Level 4), and/or full automation (Level 5), depending on the embodiment.
(50) The vehicle **500** may include components such as a chassis, a vehicle body, wheels (e.g., 2, 4, 6, 8, 18, etc.), tires, axles, and other components of a vehicle. The vehicle **500** may include a propulsion system **550**, such as an internal combustion engine, hybrid electric power plant, an all-

electric engine, and/or another propulsion system type. The propulsion system **550** may be connected to a drive train of the vehicle **500**, which may include a transmission, to enable the propulsion of the vehicle **500**. The propulsion system **550** may be controlled in response to receiving signals from the throttle/accelerator **552**.

(51) A steering system **554**, which may include a steering wheel, may be used to steer the vehicle **500** (e.g., along a desired path or route) when the propulsion system **550** is operating (e.g., when the vehicle is in motion). The steering system **554** may receive signals from a steering actuator **556**. The steering wheel may be optional for full automation (Level 5) functionality.

(52) The brake sensor system **546** may be used to operate the vehicle brakes in response to receiving signals from the brake actuators **548** and/or brake sensors.

(53) Controller(s) **536**, which may include one or more system on chips (SoCs) **504** (FIG. **5**C) and/or GPU(s), may provide signals (e.g., representative of commands) to one or more components and/or systems of the vehicle **500**. For example, the controller(s) may send signals to operate the vehicle brakes via one or more brake actuators **548**, to operate the steering system **554** via one or more steering actuators **556**, to operate the propulsion system **550** via one or more throttle/accelerators **552**. The controller(s) **536** may include one or more onboard (e.g., integrated) computing devices (e.g., supercomputers) that process sensor signals, and output operation commands (e.g., signals representing commands) to enable autonomous driving and/or to assist a human driver in driving the vehicle **500**. The controller(s) **536** may include a first controller **536** for autonomous driving functions, a second controller **536** for functional safety functions, a third controller **536** for artificial intelligence functionality (e.g., computer vision), a fourth controller **536** for infotainment functionality, a fifth controller **536** for redundancy in emergency conditions, and/or other controllers. In some examples, a single controller **536** may handle two or more of the above functionalities, two or more controllers **536** may handle a single functionality, and/or any combination thereof.

(54) The controller(s) **536** may provide the signals for controlling one or more components and/or systems of the vehicle **500** in response to sensor data received from one or more sensors (e.g., sensor inputs). The sensor data may be received from, for example and without limitation, global navigation satellite systems sensor(s) **558** (e.g., Global Positioning System sensor(s)), RADAR sensor(s) **560**, ultrasonic sensor(s) **562**, LIDAR sensor(s) **564**, inertial measurement unit (IMU) sensor(s) **566** (e.g., accelerometer(s), gyroscope(s), magnetic compass(es), magnetometer(s), etc.), microphone(s) **596**, stereo camera(s) **568**, wide-view camera(s) **570** (e.g., fisheye cameras), infrared camera(s) **572**, surround camera(s) **574** (e.g., 360 degree cameras), long-range and/or mid-range camera(s) **598**, speed sensor(s) **544** (e.g., for measuring the speed of the vehicle **500**), vibration sensor(s) **542**, steering sensor(s) **540**, brake sensor(s) (e.g., as part of the brake sensor system **546**), and/or other sensor types.

(55) One or more of the controller(s) **536** may receive inputs (e.g., represented by input data) from an instrument cluster **532** of the vehicle **500** and provide outputs (e.g., represented by output data, display data, etc.) via a human-machine interface (HMI) display **534**, an audible annunciator, a loudspeaker, and/or via other components of the vehicle **500**. The outputs may include information such as vehicle velocity, speed, time, map data (e.g., the HD map **522** of FIG. **5**C), location data (e.g., the vehicle's **500** location, such as on a map), direction, location of other vehicles (e.g., an occupancy grid), information about objects and status of objects as perceived by the controller(s) **536**, etc. For example, the HMI display **534** may display information about the presence of one or more objects (e.g., a street sign, caution sign, traffic light changing, etc.), and/or information about driving maneuvers the vehicle has made, is making, or will make (e.g., changing lanes now, taking exit **34**B in two miles, etc.).

(56) The vehicle **500** further includes a network interface **524** which may use one or more wireless antenna(s) **526** and/or modem(s) to communicate over one or more networks. For example, the network interface **524** may be capable of communication over LTE, WCDMA, UMTS, GSM,

CDMA2000, etc. The wireless antenna(s) **526** may also enable communication between objects in the environment (e.g., vehicles, mobile devices, etc.), using local area network(s), such as Bluetooth, Bluetooth LE, Z-Wave, ZigBee, etc., and/or low power wide-area network(s) (LPWANs), such as LoRaWAN, SigFox, etc.
(57) FIG. **5**B is an example of camera locations and fields of view for the example autonomous vehicle **500** of FIG. **5**A, in accordance with some embodiments of the present disclosure. The cameras and respective fields of view are one example embodiment and are not intended to be limiting. For example, additional and/or alternative cameras may be included and/or the cameras may be located at different locations on the vehicle **500**.
(58) The camera types for the cameras may include, but are not limited to, digital cameras that may be adapted for use with the components and/or systems of the vehicle **500**. The camera(s) may operate at automotive safety integrity level (ASIL) B and/or at another ASIL. The camera types may be capable of any image capture rate, such as 60 frames per second (fps), 520 fps, 240 fps, etc., depending on the embodiment. The cameras may be capable of using rolling shutters, global shutters, another type of shutter, or a combination thereof. In some examples, the color filter array may include a red clear clear clear (RCCC) color filter array, a red clear clear blue (RCCB) color filter array, a red blue green clear (RBGC) color filter array, a Foveon X3 color filter array, a Bayer sensors (RGGB) color filter array, a monochrome sensor color filter array, and/or another type of color filter array. In some embodiments, clear pixel cameras, such as cameras with an RCCC, an RCCB, and/or an RBGC color filter array, may be used in an effort to increase light sensitivity.
(59) In some examples, one or more of the camera(s) may be used to perform advanced driver assistance systems (ADAS) functions (e.g., as part of a redundant or fail-safe design). For example, a Multi-Function Mono Camera may be installed to provide functions including lane departure warning, traffic sign assist and intelligent headlamp control. One or more of the camera(s) (e.g., all of the cameras) may record and provide image data (e.g., video) simultaneously.
(60) One or more of the cameras may be mounted in a mounting assembly, such as a custom designed (3-D printed) assembly, in order to cut out stray light and reflections from within the car (e.g., reflections from the dashboard reflected in the windshield mirrors) which may interfere with the camera's image data capture abilities. With reference to wing-mirror mounting assemblies, the wing-mirror assemblies may be custom 3-D printed so that the camera mounting plate matches the shape of the wing-mirror. In some examples, the camera(s) may be integrated into the wing-mirror. For side-view cameras, the camera(s) may also be integrated within the four pillars at each corner of the cabin.
(61) Cameras with a field of view that include portions of the environment in front of the vehicle **500** (e.g., front-facing cameras) may be used for surround view, to help identify forward facing paths and obstacles, as well aid in, with the help of one or more controllers **536** and/or control SoCs, providing information critical to generating an occupancy grid and/or determining the preferred vehicle paths. Front-facing cameras may be used to perform many of the same ADAS functions as LIDAR, including emergency braking, pedestrian detection, and collision avoidance. Front-facing cameras may also be used for ADAS functions and systems including Lane Departure Warnings (“LDW”), Autonomous Cruise Control (“ACC”), and/or other functions such as traffic sign recognition.
(62) A variety of cameras may be used in a front-facing configuration, including, for example, a monocular camera platform that includes a CMOS (complementary metal oxide semiconductor) color imager. Another example may be a wide-view camera(s) **570** that may be used to perceive objects coming into view from the periphery (e.g., pedestrians, crossing traffic or bicycles). Although only one wide-view camera is illustrated in FIG. **5**B, there may any number of wide-view cameras **570** on the vehicle **500**. In addition, long-range camera(s) **598** (e.g., a long-view stereo camera pair) may be used for depth-based object detection, especially for objects for which a neural network has not yet been trained. The long-range camera(s) **598** may also be used for object

detection and classification, as well as basic object tracking.
(63) One or more stereo cameras **568** may also be included in a front-facing configuration. The stereo camera(s) **568** may include an integrated control unit comprising a scalable processing unit, which may provide a programmable logic (FPGA) and a multi-core micro-processor with an integrated CAN or Ethernet interface on a single chip. Such a unit may be used to generate a 3-D map of the vehicle's environment, including a distance estimate for all the points in the image. An alternative stereo camera(s) **568** may include a compact stereo vision sensor(s) that may include two camera lenses (one each on the left and right) and an image processing chip that may measure the distance from the vehicle to the target object and use the generated information (e.g., metadata) to activate the autonomous emergency braking and lane departure warning functions. Other types of stereo camera(s) **568** may be used in addition to, or alternatively from, those described herein.
(64) Cameras with a field of view that include portions of the environment to the side of the vehicle **500** (e.g., side-view cameras) may be used for surround view, providing information used to create and update the occupancy grid, as well as to generate side impact collision warnings. For example, surround camera(s) **574** (e.g., four surround cameras **574** as illustrated in FIG. **5**B) may be positioned to on the vehicle **500**. The surround camera(s) **574** may include wide-view camera(s) **570**, fisheye camera(s), 360 degree camera(s), and/or the like. Four example, four fisheye cameras may be positioned on the vehicle's front, rear, and sides. In an alternative arrangement, the vehicle may use three surround camera(s) **574** (e.g., left, right, and rear), and may leverage one or more other camera(s) (e.g., a forward-facing camera) as a fourth surround view camera.
(65) Cameras with a field of view that include portions of the environment to the rear of the vehicle **500** (e.g., rear-view cameras) may be used for park assistance, surround view, rear collision warnings, and creating and updating the occupancy grid. A wide variety of cameras may be used including, but not limited to, cameras that are also suitable as a front-facing camera(s) (e.g., long-range and/or mid-range camera(s) **598**, stereo camera(s) **568**), infrared camera(s) **572**, etc.), as described herein.
(66) FIG. **5**C is a block diagram of an example system architecture for the example autonomous vehicle **500** of FIG. **5**A, in accordance with some embodiments of the present disclosure. It should be understood that this and other arrangements described herein are set forth only as examples. Other arrangements and elements (e.g., machines, interfaces, functions, orders, groupings of functions, etc.) may be used in addition to or instead of those shown, and some elements may be omitted altogether. Further, many of the elements described herein are functional entities that may be implemented as discrete or distributed components or in conjunction with other components, and in any suitable combination and location. Various functions described herein as being performed by entities may be carried out by hardware, firmware, and/or software. For instance, various functions may be carried out by a processor executing instructions stored in memory.
(67) Each of the components, features, and systems of the vehicle **500** in FIG. **5**C are illustrated as being connected via bus **502**. The bus **502** may include a Controller Area Network (CAN) data interface (alternatively referred to herein as a “CAN bus”). A CAN may be a network inside the vehicle **500** used to aid in control of various features and functionality of the vehicle **500**, such as actuation of brakes, acceleration, braking, steering, windshield wipers, etc. A CAN bus may be configured to have dozens or even hundreds of nodes, each with its own unique identifier (e.g., a CAN ID). The CAN bus may be read to find steering wheel angle, ground speed, engine revolutions per minute (RPMs), button positions, and/or other vehicle status indicators. The CAN bus may be ASIL B compliant.
(68) Although the bus **502** is described herein as being a CAN bus, this is not intended to be limiting. For example, in addition to, or alternatively from, the CAN bus, FlexRay and/or Ethernet may be used. Additionally, although a single line is used to represent the bus **502**, this is not intended to be limiting. For example, there may be any number of busses **502**, which may include one or more CAN busses, one or more FlexRay busses, one or more Ethernet busses, and/or one or

more other types of busses using a different protocol. In some examples, two or more busses **502** may be used to perform different functions, and/or may be used for redundancy. For example, a first bus **502** may be used for collision avoidance functionality and a second bus **502** may be used for actuation control. In any example, each bus **502** may communicate with any of the components of the vehicle **500**, and two or more busses **502** may communicate with the same components. In some examples, each SoC **504**, each controller **536**, and/or each computer within the vehicle may have access to the same input data (e.g., inputs from sensors of the vehicle **500**), and may be connected to a common bus, such the CAN bus.

(69) The vehicle **500** may include one or more controller(s) **536**, such as those described herein with respect to FIG. **5**A. The controller(s) **536** may be used for a variety of functions. The controller(s) **536** may be coupled to any of the various other components and systems of the vehicle **500**, and may be used for control of the vehicle **500**, artificial intelligence of the vehicle **500**, infotainment for the vehicle **500**, and/or the like.

(70) The vehicle **500** may include a system(s) on a chip (SoC) **504**. The SoC **504** may include CPU(s) **506**, GPU(s) **508**, processor(s) **510**, cache(s) **512**, accelerator(s) **514**, data store(s) **516**, and/or other components and features not illustrated. The SoC(s) **504** may be used to control the vehicle **500** in a variety of platforms and systems. For example, the SoC(s) **504** may be combined in a system (e.g., the system of the vehicle **500**) with an HD map **522** which may obtain map refreshes and/or updates via a network interface **524** from one or more servers (e.g., server(s) **578** of FIG. **5**D).

(71) The CPU(s) **506** may include a CPU cluster or CPU complex (alternatively referred to herein as a "CCPLEX"). The CPU(s) **506** may include multiple cores and/or L2 caches. For example, in some embodiments, the CPU(s) **506** may include eight cores in a coherent multi-processor configuration. In some embodiments, the CPU(s) **506** may include four dual-core clusters where each cluster has a dedicated L2 cache (e.g., a 2 MB L2 cache). The CPU(s) **506** (e.g., the CCPLEX) may be configured to support simultaneous cluster operation enabling any combination of the clusters of the CPU(s) **506** to be active at any given time.

(72) The CPU(s) **506** may implement power management capabilities that include one or more of the following features: individual hardware blocks may be clock-gated automatically when idle to save dynamic power; each core clock may be gated when the core is not actively executing instructions due to execution of WFI/WFE instructions; each core may be independently power-gated; each core cluster may be independently clock-gated when all cores are clock-gated or power-gated; and/or each core cluster may be independently power-gated when all cores are power-gated. The CPU(s) **506** may further implement an enhanced algorithm for managing power states, where allowed power states and expected wakeup times are specified, and the hardware/microcode determines the best power state to enter for the core, cluster, and CCPLEX. The processing cores may support simplified power state entry sequences in software with the work offloaded to microcode.

(73) The GPU(s) **508** may include an integrated GPU (alternatively referred to herein as an "iGPU"). The GPU(s) **508** may be programmable and may be efficient for parallel workloads. The GPU(s) **508**, in some examples, may use an enhanced tensor instruction set. The GPU(s) **508** may include one or more streaming microprocessors, where each streaming microprocessor may include an L1 cache (e.g., an L1 cache with at least 96 KB storage capacity), and two or more of the streaming microprocessors may share an L2 cache (e.g., an L2 cache with a 512 KB storage capacity). In some embodiments, the GPU(s) **508** may include at least eight streaming microprocessors. The GPU(s) **508** may use compute application programming interface(s) (API(s)). In addition, the GPU(s) **508** may use one or more parallel computing platforms and/or programming models (e.g., NVIDIA's CUDA).

(74) The GPU(s) **508** may be power-optimized for best performance in automotive and embedded use cases. For example, the GPU(s) **508** may be fabricated on a Fin field-effect transistor (FinFET).

However, this is not intended to be limiting and the GPU(s) **508** may be fabricated using other semiconductor manufacturing processes. Each streaming microprocessor may incorporate a number of mixed-precision processing cores partitioned into multiple blocks. For example, and without limitation, 64 PF32 cores and 32 PF64 cores may be partitioned into four processing blocks. In such an example, each processing block may be allocated 16 FP32 cores, 8 FP64 cores, 16 INT32 cores, two mixed-precision NVIDIA TENSOR COREs for deep learning matrix arithmetic, an L0 instruction cache, a warp scheduler, a dispatch unit, and/or a 64 KB register file. In addition, the streaming microprocessors may include independent parallel integer and floating-point data paths to provide for efficient execution of workloads with a mix of computation and addressing calculations. The streaming microprocessors may include independent thread scheduling capability to enable finer-grain synchronization and cooperation between parallel threads. The streaming microprocessors may include a combined L1 data cache and shared memory unit in order to improve performance while simplifying programming.

(75) The GPU(s) **508** may include a high bandwidth memory (HBM) and/or a 16 GB HBM2 memory subsystem to provide, in some examples, about 900 GB/second peak memory bandwidth. In some examples, in addition to, or alternatively from, the HBM memory, a synchronous graphics random-access memory (SGRAM) may be used, such as a graphics double data rate type five synchronous random-access memory (GDDR5).

(76) The GPU(s) **508** may include unified memory technology including access counters to allow for more accurate migration of memory pages to the processor that accesses them most frequently, thereby improving efficiency for memory ranges shared between processors. In some examples, address translation services (ATS) support may be used to allow the GPU(s) **508** to access the CPU(s) **506** page tables directly. In such examples, when the GPU(s) **508** memory management unit (MMU) experiences a miss, an address translation request may be transmitted to the CPU(s) **506**. In response, the CPU(s) **506** may look in its page tables for the virtual-to-physical mapping for the address and transmits the translation back to the GPU(s) **508**. As such, unified memory technology may allow a single unified virtual address space for memory of both the CPU(s) **506** and the GPU(s) **508**, thereby simplifying the GPU(s) **508** programming and porting of applications to the GPU(s) **508**.

(77) In addition, the GPU(s) **508** may include an access counter that may keep track of the frequency of access of the GPU(s) **508** to memory of other processors. The access counter may help ensure that memory pages are moved to the physical memory of the processor that is accessing the pages most frequently.

(78) The SoC(s) **504** may include any number of cache(s) **512**, including those described herein. For example, the cache(s) **512** may include an L3 cache that is available to both the CPU(s) **506** and the GPU(s) **508** (e.g., that is connected both the CPU(s) **506** and the GPU(s) **508**). The cache(s) **512** may include a write-back cache that may keep track of states of lines, such as by using a cache coherence protocol (e.g., MEI, MESI, MSI, etc.). The L3 cache may include 4 MB or more, depending on the embodiment, although smaller cache sizes may be used.

(79) The SoC(s) **504** may include an arithmetic logic unit(s) (ALU(s)) which may be leveraged in performing processing with respect to any of the variety of tasks or operations of the vehicle **500**—such as processing DNNs. In addition, the SoC(s) **504** may include a floating point unit(s) (FPU(s))—or other math coprocessor or numeric coprocessor types—for performing mathematical operations within the system. For example, the SoC(s) **504** may include one or more FPUs integrated as execution units within a CPU(s) **506** and/or GPU(s) **508**.

(80) The SoC(s) **504** may include one or more accelerators **514** (e.g., hardware accelerators, software accelerators, or a combination thereof). For example, the SoC(s) **504** may include a hardware acceleration cluster that may include optimized hardware accelerators and/or large on-chip memory. The large on-chip memory (e.g., 4 MB of SRAM), may enable the hardware acceleration cluster to accelerate neural networks and other calculations. The hardware acceleration

cluster may be used to complement the GPU(s) **508** and to off-load some of the tasks of the GPU(s) **508** (e.g., to free up more cycles of the GPU(s) **508** for performing other tasks). As an example, the accelerator(s) **514** may be used for targeted workloads (e.g., perception, convolutional neural networks (CNNs), etc.) that are stable enough to be amenable to acceleration. The term "CNN," as used herein, may include all types of CNNs, including region-based or regional convolutional neural networks (RCNNs) and Fast RCNNs (e.g., as used for object detection).

(81) The accelerator(s) **514** (e.g., the hardware acceleration cluster) may include a deep learning accelerator(s) (DLA). The DLA(s) may include one or more Tensor processing units (TPUs) that may be configured to provide an additional ten trillion operations per second for deep learning applications and inferencing. The TPUs may be accelerators configured to, and optimized for, performing image processing functions (e.g., for CNNs, RCNNs, etc.). The DLA(s) may further be optimized for a specific set of neural network types and floating point operations, as well as inferencing. The design of the DLA(s) may provide more performance per millimeter than a general-purpose GPU, and vastly exceeds the performance of a CPU. The TPU(s) may perform several functions, including a single-instance convolution function, supporting, for example, INT8, INT16, and FP16 data types for both features and weights, as well as post-processor functions.

(82) The DLA(s) may quickly and efficiently execute neural networks, especially CNNs, on processed or unprocessed data for any of a variety of functions, including, for example and without limitation: a CNN for object identification and detection using data from camera sensors; a CNN for distance estimation using data from camera sensors; a CNN for emergency vehicle detection and identification and detection using data from microphones; a CNN for facial recognition and vehicle owner identification using data from camera sensors; and/or a CNN for security and/or safety related events.

(83) The DLA(s) may perform any function of the GPU(s) **508**, and by using an inference accelerator, for example, a designer may target either the DLA(s) or the GPU(s) **508** for any function. For example, the designer may focus processing of CNNs and floating point operations on the DLA(s) and leave other functions to the GPU(s) **508** and/or other accelerator(s) **514**.

(84) The accelerator(s) **514** (e.g., the hardware acceleration cluster) may include a programmable vision accelerator(s) (PVA), which may alternatively be referred to herein as a computer vision accelerator. The PVA(s) may be designed and configured to accelerate computer vision algorithms for the advanced driver assistance systems (ADAS), autonomous driving, and/or augmented reality (AR) and/or virtual reality (VR) applications. The PVA(s) may provide a balance between performance and flexibility. For example, each PVA(s) may include, for example and without limitation, any number of reduced instruction set computer (RISC) cores, direct memory access (DMA), and/or any number of vector processors.

(85) The RISC cores may interact with image sensors (e.g., the image sensors of any of the cameras described herein), image signal processor(s), and/or the like. Each of the RISC cores may include any amount of memory. The RISC cores may use any of a number of protocols, depending on the embodiment. In some examples, the RISC cores may execute a real-time operating system (RTOS). The RISC cores may be implemented using one or more integrated circuit devices, application specific integrated circuits (ASICs), and/or memory devices. For example, the RISC cores may include an instruction cache and/or a tightly coupled RAM.

(86) The DMA may enable components of the PVA(s) to access the system memory independently of the CPU(s) **506**. The DMA may support any number of features used to provide optimization to the PVA including, but not limited to, supporting multi-dimensional addressing and/or circular addressing. In some examples, the DMA may support up to six or more dimensions of addressing, which may include block width, block height, block depth, horizontal block stepping, vertical block stepping, and/or depth stepping.

(87) The vector processors may be programmable processors that may be designed to efficiently and flexibly execute programming for computer vision algorithms and provide signal processing

capabilities. In some examples, the PVA may include a PVA core and two vector processing subsystem partitions. The PVA core may include a processor subsystem, DMA engine(s) (e.g., two DMA engines), and/or other peripherals. The vector processing subsystem may operate as the primary processing engine of the PVA, and may include a vector processing unit (VPU), an instruction cache, and/or vector memory (e.g., VMEM). A VPU core may include a digital signal processor such as, for example, a single instruction, multiple data (SIMD), very long instruction word (VLIW) digital signal processor. The combination of the SIMD and VLIW may enhance throughput and speed.

(88) Each of the vector processors may include an instruction cache and may be coupled to dedicated memory. As a result, in some examples, each of the vector processors may be configured to execute independently of the other vector processors. In other examples, the vector processors that are included in a particular PVA may be configured to employ data parallelism. For example, in some embodiments, the plurality of vector processors included in a single PVA may execute the same computer vision algorithm, but on different regions of an image. In other examples, the vector processors included in a particular PVA may simultaneously execute different computer vision algorithms, on the same image, or even execute different algorithms on sequential images or portions of an image. Among other things, any number of PVAs may be included in the hardware acceleration cluster and any number of vector processors may be included in each of the PVAs. In addition, the PVA(s) may include additional error correcting code (ECC) memory, to enhance overall system safety.

(89) The accelerator(s) **514** (e.g., the hardware acceleration cluster) may include a computer vision network on-chip and SRAM, for providing a high-bandwidth, low latency SRAM for the accelerator(s) **514**. In some examples, the on-chip memory may include at least 4 MB SRAM, consisting of, for example and without limitation, eight field-configurable memory blocks, that may be accessible by both the PVA and the DLA. Each pair of memory blocks may include an advanced peripheral bus (APB) interface, configuration circuitry, a controller, and a multiplexer. Any type of memory may be used. The PVA and DLA may access the memory via a backbone that provides the PVA and DLA with high-speed access to memory. The backbone may include a computer vision network on-chip that interconnects the PVA and the DLA to the memory (e.g., using the APB).

(90) The computer vision network on-chip may include an interface that determines, before transmission of any control signal/address/data, that both the PVA and the DLA provide ready and valid signals. Such an interface may provide for separate phases and separate channels for transmitting control signals/addresses/data, as well as burst-type communications for continuous data transfer. This type of interface may comply with ISO 26262 or IEC 61508 standards, although other standards and protocols may be used.

(91) In some examples, the SoC(s) **504** may include a real-time ray-tracing hardware accelerator, such as described in U.S. patent application Ser. No. 16/101,232, filed on Aug. 10, 2018. The real-time ray-tracing hardware accelerator may be used to quickly and efficiently determine the positions and extents of objects (e.g., within a world model), to generate real-time visualization simulations, for RADAR signal interpretation, for sound propagation synthesis and/or analysis, for simulation of SONAR systems, for general wave propagation simulation, for comparison to LIDAR data for purposes of localization and/or other functions, and/or for other uses. In some embodiments, one or more tree traversal units (TTUs) may be used for executing one or more ray-tracing related operations.

(92) The accelerator(s) **514** (e.g., the hardware accelerator cluster) have a wide array of uses for autonomous driving. The PVA may be a programmable vision accelerator that may be used for key processing stages in ADAS and autonomous vehicles. The PVA's capabilities are a good match for algorithmic domains needing predictable processing, at low power and low latency. In other words, the PVA performs well on semi-dense or dense regular computation, even on small data sets, which

need predictable run-times with low latency and low power. Thus, in the context of platforms for autonomous vehicles, the PVAs are designed to run classic computer vision algorithms, as they are efficient at object detection and operating on integer math.

(93) For example, according to one embodiment of the technology, the PVA is used to perform computer stereo vision. A semi-global matching-based algorithm may be used in some examples, although this is not intended to be limiting. Many applications for Level 3-5 autonomous driving require motion estimation/stereo matching on-the-fly (e.g., structure from motion, pedestrian recognition, lane detection, etc.). The PVA may perform computer stereo vision function on inputs from two monocular cameras.

(94) In some examples, the PVA may be used to perform dense optical flow. According to process raw RADAR data (e.g., using a 4D Fast Fourier Transform) to provide Processed RADAR. In other examples, the PVA is used for time of flight depth processing, by processing raw time of flight data to provide processed time of flight data, for example.

(95) The DLA may be used to run any type of network to enhance control and driving safety, including for example, a neural network that outputs a measure of confidence for each object detection. Such a confidence value may be interpreted as a probability, or as providing a relative "weight" of each detection compared to other detections. This confidence value enables the system to make further decisions regarding which detections should be considered as true positive detections rather than false positive detections. For example, the system may set a threshold value for the confidence and consider only the detections exceeding the threshold value as true positive detections. In an automatic emergency braking (AEB) system, false positive detections would cause the vehicle to automatically perform emergency braking, which is obviously undesirable. Therefore, only the most confident detections should be considered as triggers for AEB. The DLA may run a neural network for regressing the confidence value. The neural network may take as its input at least some subset of parameters, such as bounding box dimensions, ground plane estimate obtained (e.g. from another subsystem), inertial measurement unit (IMU) sensor **566** output that correlates with the vehicle **500** orientation, distance, 3D location estimates of the object obtained from the neural network and/or other sensors (e.g., LIDAR sensor(s) **564** or RADAR sensor(s) **560**), among others.

(96) The SoC(s) **504** may include data store(s) **516** (e.g., memory). The data store(s) **516** may be on-chip memory of the SoC(s) **504**, which may store neural networks to be executed on the GPU and/or the DLA. In some examples, the data store(s) **516** may be large enough in capacity to store multiple instances of neural networks for redundancy and safety. The data store(s) **512** may comprise L2 or L3 cache(s) **512**. Reference to the data store(s) **516** may include reference to the memory associated with the PVA, DLA, and/or other accelerator(s) **514**, as described herein.

(97) The SoC(s) **504** may include one or more processor(s) **510** (e.g., embedded processors). The processor(s) **510** may include a boot and power management processor that may be a dedicated processor and subsystem to handle boot power and management functions and related security enforcement. The boot and power management processor may be a part of the SoC(s) **504** boot sequence and may provide runtime power management services. The boot power and management processor may provide clock and voltage programming, assistance in system low power state transitions, management of SoC(s) **504** thermals and temperature sensors, and/or management of the SoC(s) **504** power states. Each temperature sensor may be implemented as a ring-oscillator whose output frequency is proportional to temperature, and the SoC(s) **504** may use the ring-oscillators to detect temperatures of the CPU(s) **506**, GPU(s) **508**, and/or accelerator(s) **514**. If temperatures are determined to exceed a threshold, the boot and power management processor may enter a temperature fault routine and put the SoC(s) **504** into a lower power state and/or put the vehicle **500** into a chauffeur to safe stop mode (e.g., bring the vehicle **500** to a safe stop).

(98) The processor(s) **510** may further include a set of embedded processors that may serve as an audio processing engine. The audio processing engine may be an audio subsystem that enables full

hardware support for multi-channel audio over multiple interfaces, and a broad and flexible range of audio I/O interfaces. In some examples, the audio processing engine is a dedicated processor core with a digital signal processor with dedicated RAM.
(99) The processor(s) **510** may further include an always on processor engine that may provide necessary hardware features to support low power sensor management and wake use cases. The always on processor engine may include a processor core, a tightly coupled RAM, supporting peripherals (e.g., timers and interrupt controllers), various I/O controller peripherals, and routing logic.
(100) The processor(s) **510** may further include a safety cluster engine that includes a dedicated processor subsystem to handle safety management for automotive applications. The safety cluster engine may include two or more processor cores, a tightly coupled RAM, support peripherals (e.g., timers, an interrupt controller, etc.), and/or routing logic. In a safety mode, the two or more cores may operate in a lockstep mode and function as a single core with comparison logic to detect any differences between their operations.
(101) The processor(s) **510** may further include a real-time camera engine that may include a dedicated processor subsystem for handling real-time camera management.
(102) The processor(s) **510** may further include a high-dynamic range signal processor that may include an image signal processor that is a hardware engine that is part of the camera processing pipeline.
(103) The processor(s) **510** may include a video image compositor that may be a processing block (e.g., implemented on a microprocessor) that implements video post-processing functions needed by a video playback application to produce the final image for the player window. The video image compositor may perform lens distortion correction on wide-view camera(s) **570**, surround camera(s) **574**, and/or on in-cabin monitoring camera sensors. In-cabin monitoring camera sensor is preferably monitored by a neural network running on another instance of the Advanced SoC, configured to identify in cabin events and respond accordingly. An in-cabin system may perform lip reading to activate cellular service and place a phone call, dictate emails, change the vehicle's destination, activate or change the vehicle's infotainment system and settings, or provide voice-activated web surfing. Certain functions are available to the driver only when the vehicle is operating in an autonomous mode, and are disabled otherwise.
(104) The video image compositor may include enhanced temporal noise reduction for both spatial and temporal noise reduction. For example, where motion occurs in a video, the noise reduction weights spatial information appropriately, decreasing the weight of information provided by adjacent frames. Where an image or portion of an image does not include motion, the temporal noise reduction performed by the video image compositor may use information from the previous image to reduce noise in the current image.
(105) The video image compositor may also be configured to perform stereo rectification on input stereo lens frames. The video image compositor may further be used for user interface composition when the operating system desktop is in use, and the GPU(s) **508** is not required to continuously render new surfaces. Even when the GPU(s) **508** is powered on and active doing 3D rendering, the video image compositor may be used to offload the GPU(s) **508** to improve performance and responsiveness.
(106) The SoC(s) **504** may further include a mobile industry processor interface (MIPI) camera serial interface for receiving video and input from cameras, a high-speed interface, and/or a video input block that may be used for camera and related pixel input functions. The SoC(s) **504** may further include an input/output controller(s) that may be controlled by software and may be used for receiving I/O signals that are uncommitted to a specific role.
(107) The SoC(s) **504** may further include a broad range of peripheral interfaces to enable communication with peripherals, audio codecs, power management, and/or other devices. The SoC(s) **504** may be used to process data from cameras (e.g., connected over Gigabit Multimedia

Serial Link and Ethernet), sensors (e.g., LIDAR sensor(s) **564**, RADAR sensor(s) **560**, etc. that may be connected over Ethernet), data from bus **502** (e.g., speed of vehicle **500**, steering wheel position, etc.), data from GNSS sensor(s) **558** (e.g., connected over Ethernet or CAN bus). The SoC(s) **504** may further include dedicated high-performance mass storage controllers that may include their own DMA engines, and that may be used to free the CPU(s) **506** from routine data management tasks.

(108) The SoC(s) **504** may be an end-to-end platform with a flexible architecture that spans automation levels 3-5, thereby providing a comprehensive functional safety architecture that leverages and makes efficient use of computer vision and ADAS techniques for diversity and redundancy, provides a platform for a flexible, reliable driving software stack, along with deep learning tools. The SoC(s) **504** may be faster, more reliable, and even more energy-efficient and space-efficient than conventional systems. For example, the accelerator(s) **514**, when combined with the CPU(s) **506**, the GPU(s) **508**, and the data store(s) **516**, may provide for a fast, efficient platform for level 3-5 autonomous vehicles.

(109) The technology thus provides capabilities and functionality that cannot be achieved by conventional systems. For example, computer vision algorithms may be executed on CPUs, which may be configured using high-level programming language, such as the C programming language, to execute a wide variety of processing algorithms across a wide variety of visual data. However, CPUs are oftentimes unable to meet the performance requirements of many computer vision applications, such as those related to execution time and power consumption, for example. In particular, many CPUs are unable to execute complex object detection algorithms in real-time, which is a requirement of in-vehicle ADAS applications, and a requirement for practical Level 3-5 autonomous vehicles.

(110) In contrast to conventional systems, by providing a CPU complex, GPU complex, and a hardware acceleration cluster, the technology described herein allows for multiple neural networks to be performed simultaneously and/or sequentially, and for the results to be combined together to enable Level 3-5 autonomous driving functionality. For example, a CNN executing on the DLA or dGPU (e.g., the GPU(s) **520**) may include a text and word recognition, allowing the supercomputer to read and understand traffic signs, including signs for which the neural network has not been specifically trained. The DLA may further include a neural network that is able to identify, interpret, and provides semantic understanding of the sign, and to pass that semantic understanding to the path planning modules running on the CPU Complex.

(111) As another example, multiple neural networks may be run simultaneously, as is required for Level 3, 4, or 5 driving. For example, a warning sign consisting of “Caution: flashing lights indicate icy conditions,” along with an electric light, may be independently or collectively interpreted by several neural networks. The sign itself may be identified as a traffic sign by a first deployed neural network (e.g., a neural network that has been trained), the text “Flashing lights indicate icy conditions” may be interpreted by a second deployed neural network, which informs the vehicle's path planning software (preferably executing on the CPU Complex) that when flashing lights are detected, icy conditions exist. The flashing light may be identified by operating a third deployed neural network over multiple frames, informing the vehicle's path-planning software of the presence (or absence) of flashing lights. All three neural networks may run simultaneously, such as within the DLA and/or on the GPU(s) **508**.

(112) In some examples, a CNN for facial recognition and vehicle owner identification may use data from camera sensors to identify the presence of an authorized driver and/or owner of the vehicle **500**. The always on sensor processing engine may be used to unlock the vehicle when the owner approaches the driver door and turn on the lights, and, in security mode, to disable the vehicle when the owner leaves the vehicle. In this way, the SoC(s) **504** provide for security against theft and/or carjacking.

(113) In another example, a CNN for emergency vehicle detection and identification may use data

from microphones **596** to detect and identify emergency vehicle sirens. In contrast to conventional systems, that use general classifiers to detect sirens and manually extract features, the SoC(s) **504** use the CNN for classifying environmental and urban sounds, as well as classifying visual data. In a preferred embodiment, the CNN running on the DLA is trained to identify the relative closing speed of the emergency vehicle (e.g., by using the Doppler Effect). The CNN may also be trained to identify emergency vehicles specific to the local area in which the vehicle is operating, as identified by GNSS sensor(s) **558**. Thus, for example, when operating in Europe the CNN will seek to detect European sirens, and when in the United States the CNN will seek to identify only North American sirens. Once an emergency vehicle is detected, a control program may be used to execute an emergency vehicle safety routine, slowing the vehicle, pulling over to the side of the road, parking the vehicle, and/or idling the vehicle, with the assistance of ultrasonic sensors **562**, until the emergency vehicle(s) passes.

(114) The vehicle may include a CPU(s) **518** (e.g., discrete CPU(s), or dCPU(s)), that may be coupled to the SoC(s) **504** via a high-speed interconnect (e.g., PCIe). The CPU(s) **518** may include an X86 processor, for example. The CPU(s) **518** may be used to perform any of a variety of functions, including arbitrating potentially inconsistent results between ADAS sensors and the SoC(s) **504**, and/or monitoring the status and health of the controller(s) **536** and/or infotainment SoC **530**, for example.

(115) The vehicle **500** may include a GPU(s) **520** (e.g., discrete GPU(s), or dGPU(s)), that may be coupled to the SoC(s) **504** via a high-speed interconnect (e.g., NVIDIA's NVLINK). The GPU(s) **520** may provide additional artificial intelligence functionality, such as by executing redundant and/or different neural networks, and may be used to train and/or update neural networks based on input (e.g., sensor data) from sensors of the vehicle **500**.

(116) The vehicle **500** may further include the network interface **524** which may include one or more wireless antennas **526** (e.g., one or more wireless antennas for different communication protocols, such as a cellular antenna, a Bluetooth antenna, etc.). The network interface **524** may be used to enable wireless connectivity over the Internet with the cloud (e.g., with the server(s) **578** and/or other network devices), with other vehicles, and/or with computing devices (e.g., client devices of passengers). To communicate with other vehicles, a direct link may be established between the two vehicles and/or an indirect link may be established (e.g., across networks and over the Internet). Direct links may be provided using a vehicle-to-vehicle communication link. The vehicle-to-vehicle communication link may provide the vehicle **500** information about vehicles in proximity to the vehicle **500** (e.g., vehicles in front of, on the side of, and/or behind the vehicle **500**). This functionality may be part of a cooperative adaptive cruise control functionality of the vehicle **500**.

(117) The network interface **524** may include a SoC that provides modulation and demodulation functionality and enables the controller(s) **536** to communicate over wireless networks. The network interface **524** may include a radio frequency front-end for up-conversion from baseband to radio frequency, and down conversion from radio frequency to baseband. The frequency conversions may be performed through well-known processes, and/or may be performed using super-heterodyne processes. In some examples, the radio frequency front end functionality may be provided by a separate chip. The network interface may include wireless functionality for communicating over LTE, WCDMA, UMTS, GSM, CDMA2000, Bluetooth, Bluetooth LE, Wi-Fi, Z-Wave, ZigBee, LoRaWAN, and/or other wireless protocols.

(118) The vehicle **500** may further include data store(s) **528** which may include off-chip (e.g., off the SoC(s) **504**) storage. The data store(s) **528** may include one or more storage elements including RAM, SRAM, DRAM, VRAM, Flash, hard disks, and/or other components and/or devices that may store at least one bit of data.

(119) The vehicle **500** may further include GNSS sensor(s) **558**. The GNSS sensor(s) **558** (e.g., GPS and/or assisted GPS sensors), to assist in mapping, perception, occupancy grid generation,

and/or path planning functions. Any number of GNSS sensor(s) **558** may be used, including, for example and without limitation, a GPS using a USB connector with an Ethernet to Serial (RS-232) bridge.

(120) The vehicle **500** may further include RADAR sensor(s) **560**. The RADAR sensor(s) **560** may be used by the vehicle **500** for long-range vehicle detection, even in darkness and/or severe weather conditions. RADAR functional safety levels may be ASIL B. The RADAR sensor(s) **560** may use the CAN and/or the bus **502** (e.g., to transmit data generated by the RADAR sensor(s) **560**) for control and to access object tracking data, with access to Ethernet to access raw data in some examples. A wide variety of RADAR sensor types may be used. For example, and without limitation, the RADAR sensor(s) **560** may be suitable for front, rear, and side RADAR use. In some example, Pulse Doppler RADAR sensor(s) are used.

(121) The RADAR sensor(s) **560** may include different configurations, such as long range with narrow field of view, short range with wide field of view, short range side coverage, etc. In some examples, long-range RADAR may be used for adaptive cruise control functionality. The long-range RADAR systems may provide a broad field of view realized by two or more independent scans, such as within a 250 m range. The RADAR sensor(s) **560** may help in distinguishing between static and moving objects, and may be used by ADAS systems for emergency brake assist and forward collision warning. Long-range RADAR sensors may include monostatic multimodal RADAR with multiple (e.g., six or more) fixed RADAR antennae and a high-speed CAN and FlexRay interface. In an example with six antennae, the central four antennae may create a focused beam pattern, designed to record the vehicle's **500** surroundings at higher speeds with minimal interference from traffic in adjacent lanes. The other two antennae may expand the field of view, making it possible to quickly detect vehicles entering or leaving the vehicle's **500** lane.

(122) Mid-range RADAR systems may include, as an example, a range of up to 560 m (front) or 80 m (rear), and a field of view of up to 42 degrees (front) or 550 degrees (rear). Short-range RADAR systems may include, without limitation, RADAR sensors designed to be installed at both ends of the rear bumper. When installed at both ends of the rear bumper, such a RADAR sensor systems may create two beams that constantly monitor the blind spot in the rear and next to the vehicle.

(123) Short-range RADAR systems may be used in an ADAS system for blind spot detection and/or lane change assist.

(124) The vehicle **500** may further include ultrasonic sensor(s) **562**. The ultrasonic sensor(s) **562**, which may be positioned at the front, back, and/or the sides of the vehicle **500**, may be used for park assist and/or to create and update an occupancy grid. A wide variety of ultrasonic sensor(s) **562** may be used, and different ultrasonic sensor(s) **562** may be used for different ranges of detection (e.g., 2.5 m, 4 m). The ultrasonic sensor(s) **562** may operate at functional safety levels of ASIL B.

(125) The vehicle **500** may include LIDAR sensor(s) **564**. The LIDAR sensor(s) **564** may be used for object and pedestrian detection, emergency braking, collision avoidance, and/or other functions. The LIDAR sensor(s) **564** may be functional safety level ASIL B. In some examples, the vehicle **500** may include multiple LIDAR sensors **564** (e.g., two, four, six, etc.) that may use Ethernet (e.g., to provide data to a Gigabit Ethernet switch).

(126) In some examples, the LIDAR sensor(s) **564** may be capable of providing a list of objects and their distances for a 360-degree field of view. Commercially available LIDAR sensor(s) **564** may have an advertised range of approximately 500 m, with an accuracy of 2 cm-3 cm, and with support for a 500 Mbps Ethernet connection, for example. In some examples, one or more non-protruding LIDAR sensors **564** may be used. In such examples, the LIDAR sensor(s) **564** may be implemented as a small device that may be embedded into the front, rear, sides, and/or corners of the vehicle **500**. The LIDAR sensor(s) **564**, in such examples, may provide up to a 520-degree horizontal and 35-degree vertical field-of-view, with a 200 m range even for low-reflectivity objects. Front-mounted LIDAR sensor(s) **564** may be configured for a horizontal field of view

between 45 degrees and 135 degrees.

(127) In some examples, LIDAR technologies, such as 3D flash LIDAR, may also be used. 3D Flash LIDAR uses a flash of a laser as a transmission source, to illuminate vehicle surroundings up to approximately 200 m. A flash LIDAR unit includes a receptor, which records the laser pulse transit time and the reflected light on each pixel, which in turn corresponds to the range from the vehicle to the objects. Flash LIDAR may allow for highly accurate and distortion-free images of the surroundings to be generated with every laser flash. In some examples, four flash LIDAR sensors may be deployed, one at each side of the vehicle **500**. Available 3D flash LIDAR systems include a solid-state 3D staring array LIDAR camera with no moving parts other than a fan (e.g., a non-scanning LIDAR device). The flash LIDAR device may use a 5 nanosecond class I (eye-safe) laser pulse per frame and may capture the reflected laser light in the form of 3D range point clouds and co-registered intensity data. By using flash LIDAR, and because flash LIDAR is a solid-state device with no moving parts, the LIDAR sensor(s) **564** may be less susceptible to motion blur, vibration, and/or shock.

(128) The vehicle may further include IMU sensor(s) **566**. The IMU sensor(s) **566** may be located at a center of the rear axle of the vehicle **500**, in some examples. The IMU sensor(s) **566** may include, for example and without limitation, an accelerometer(s), a magnetometer(s), a gyroscope(s), a magnetic compass(es), and/or other sensor types. In some examples, such as in six-axis applications, the IMU sensor(s) **566** may include accelerometers and gyroscopes, while in nine-axis applications, the IMU sensor(s) **566** may include accelerometers, gyroscopes, and magnetometers.

(129) In some embodiments, the IMU sensor(s) **566** may be implemented as a miniature, high performance GPS-Aided Inertial Navigation System (GPS/INS) that combines micro-electro-mechanical systems (MEMS) inertial sensors, a high-sensitivity GPS receiver, and advanced Kalman filtering algorithms to provide estimates of position, velocity, and attitude. As such, in some examples, the IMU sensor(s) **566** may enable the vehicle **500** to estimate heading without requiring input from a magnetic sensor by directly observing and correlating the changes in velocity from GPS to the IMU sensor(s) **566**. In some examples, the IMU sensor(s) **566** and the GNSS sensor(s) **558** may be combined in a single integrated unit.

(130) The vehicle may include microphone(s) **596** placed in and/or around the vehicle **500**. The microphone(s) **596** may be used for emergency vehicle detection and identification, among other things.

(131) The vehicle may further include any number of camera types, including stereo camera(s) **568**, wide-view camera(s) **570**, infrared camera(s) **572**, surround camera(s) **574**, long-range and/or mid-range camera(s) **598**, and/or other camera types. The cameras may be used to capture image data around an entire periphery of the vehicle **500**. The types of cameras used depends on the embodiments and requirements for the vehicle **500**, and any combination of camera types may be used to provide the necessary coverage around the vehicle **500**. In addition, the number of cameras may differ depending on the embodiment. For example, the vehicle may include six cameras, seven cameras, ten cameras, twelve cameras, and/or another number of cameras. The cameras may support, as an example and without limitation, Gigabit Multimedia Serial Link (GMSL) and/or Gigabit Ethernet. Each of the camera(s) is described with more detail herein with respect to FIG. 5A and FIG. 5B.

(132) The vehicle **500** may further include vibration sensor(s) **542**. The vibration sensor(s) **542** may measure vibrations of components of the vehicle, such as the axle(s). For example, changes in vibrations may indicate a change in road surfaces. In another example, when two or more vibration sensors **542** are used, the differences between the vibrations may be used to determine friction or slippage of the road surface (e.g., when the difference in vibration is between a power-driven axle and a freely rotating axle).

(133) The vehicle **500** may include an ADAS system **538**. The ADAS system **538** may include a

SoC, in some examples. The ADAS system **538** may include autonomous/adaptive/automatic cruise control (ACC), cooperative adaptive cruise control (CACC), forward crash warning (FCW), automatic emergency braking (AEB), lane departure warnings (LDW), lane keep assist (LKA), blind spot warning (BSW), rear cross-traffic warning (RCTW), collision warning systems (CWS), lane centering (LC), and/or other features and functionality.

(134) The ACC systems may use RADAR sensor(s) **560**, LIDAR sensor(s) **564**, and/or a camera(s). The ACC systems may include longitudinal ACC and/or lateral ACC. Longitudinal ACC monitors and controls the distance to the vehicle immediately ahead of the vehicle **500** and automatically adjust the vehicle speed to maintain a safe distance from vehicles ahead. Lateral ACC performs distance keeping, and advises the vehicle **500** to change lanes when necessary. Lateral ACC is related to other ADAS applications such as LCA and CWS.

(135) CACC uses information from other vehicles that may be received via the network interface **524** and/or the wireless antenna(s) **526** from other vehicles via a wireless link, or indirectly, over a network connection (e.g., over the Internet). Direct links may be provided by a vehicle-to-vehicle (V2V) communication link, while indirect links may be infrastructure-to-vehicle (I2V) communication link. In general, the V2V communication concept provides information about the immediately preceding vehicles (e.g., vehicles immediately ahead of and in the same lane as the vehicle **500**), while the I2V communication concept provides information about traffic further ahead. CACC systems may include either or both I2V and V2V information sources. Given the information of the vehicles ahead of the vehicle **500**, CACC may be more reliable and it has potential to improve traffic flow smoothness and reduce congestion on the road.

(136) FCW systems are designed to alert the driver to a hazard, so that the driver may take corrective action. FCW systems use a front-facing camera and/or RADAR sensor(s) **560**, coupled to a dedicated processor, DSP, FPGA, and/or ASIC, that is electrically coupled to driver feedback, such as a display, speaker, and/or vibrating component. FCW systems may provide a warning, such as in the form of a sound, visual warning, vibration and/or a quick brake pulse.

(137) AEB systems detect an impending forward collision with another vehicle or other object, and may automatically apply the brakes if the driver does not take corrective action within a specified time or distance parameter. AEB systems may use front-facing camera(s) and/or RADAR sensor(s) **560**, coupled to a dedicated processor, DSP, FPGA, and/or ASIC. When the AEB system detects a hazard, it typically first alerts the driver to take corrective action to avoid the collision and, if the driver does not take corrective action, the AEB system may automatically apply the brakes in an effort to prevent, or at least mitigate, the impact of the predicted collision. AEB systems, may include techniques such as dynamic brake support and/or crash imminent braking.

(138) LDW systems provide visual, audible, and/or tactile warnings, such as steering wheel or seat vibrations, to alert the driver when the vehicle **500** crosses lane markings. A LDW system does not activate when the driver indicates an intentional lane departure, by activating a turn signal. LDW systems may use front-side facing cameras, coupled to a dedicated processor, DSP, FPGA, and/or ASIC, that is electrically coupled to driver feedback, such as a display, speaker, and/or vibrating component.

(139) LKA systems are a variation of LDW systems. LKA systems provide steering input or braking to correct the vehicle **500** if the vehicle **500** starts to exit the lane.

(140) BSW systems detects and warn the driver of vehicles in an automobile's blind spot. BSW systems may provide a visual, audible, and/or tactile alert to indicate that merging or changing lanes is unsafe. The system may provide an additional warning when the driver uses a turn signal. BSW systems may use rear-side facing camera(s) and/or RADAR sensor(s) **560**, coupled to a dedicated processor, DSP, FPGA, and/or ASIC, that is electrically coupled to driver feedback, such as a display, speaker, and/or vibrating component.

(141) RCTW systems may provide visual, audible, and/or tactile notification when an object is detected outside the rear-camera range when the vehicle **500** is backing up. Some RCTW systems

include AEB to ensure that the vehicle brakes are applied to avoid a crash. RCTW systems may use one or more rear-facing RADAR sensor(s) **560**, coupled to a dedicated processor, DSP, FPGA, and/or ASIC, that is electrically coupled to driver feedback, such as a display, speaker, and/or vibrating component.

(142) Conventional ADAS systems may be prone to false positive results which may be annoying and distracting to a driver, but typically are not catastrophic, because the ADAS systems alert the driver and allow the driver to decide whether a safety condition truly exists and act accordingly. However, in an autonomous vehicle **500**, the vehicle **500** itself must, in the case of conflicting results, decide whether to heed the result from a primary computer or a secondary computer (e.g., a first controller **536** or a second controller **536**). For example, in some embodiments, the ADAS system **538** may be a backup and/or secondary computer for providing perception information to a backup computer rationality module. The backup computer rationality monitor may run a redundant diverse software on hardware components to detect faults in perception and dynamic driving tasks. Outputs from the ADAS system **538** may be provided to a supervisory MCU. If outputs from the primary computer and the secondary computer conflict, the supervisory MCU must determine how to reconcile the conflict to ensure safe operation.

(143) In some examples, the primary computer may be configured to provide the supervisory MCU with a confidence score, indicating the primary computer's confidence in the chosen result. If the confidence score exceeds a threshold, the supervisory MCU may follow the primary computer's direction, regardless of whether the secondary computer provides a conflicting or inconsistent result. Where the confidence score does not meet the threshold, and where the primary and secondary computer indicate different results (e.g., the conflict), the supervisory MCU may arbitrate between the computers to determine the appropriate outcome.

(144) The supervisory MCU may be configured to run a neural network(s) that is trained and configured to determine, based on outputs from the primary computer and the secondary computer, conditions under which the secondary computer provides false alarms. Thus, the neural network(s) in the supervisory MCU may learn when the secondary computer's output may be trusted, and when it cannot. For example, when the secondary computer is a RADAR-based FCW system, a neural network(s) in the supervisory MCU may learn when the FCW system is identifying metallic objects that are not, in fact, hazards, such as a drainage grate or manhole cover that triggers an alarm. Similarly, when the secondary computer is a camera-based LDW system, a neural network in the supervisory MCU may learn to override the LDW when bicyclists or pedestrians are present and a lane departure is, in fact, the safest maneuver. In embodiments that include a neural network(s) running on the supervisory MCU, the supervisory MCU may include at least one of a DLA or GPU suitable for running the neural network(s) with associated memory. In preferred embodiments, the supervisory MCU may comprise and/or be included as a component of the SoC(s) **504**.

(145) In other examples, ADAS system **538** may include a secondary computer that performs ADAS functionality using traditional rules of computer vision. As such, the secondary computer may use classic computer vision rules (if-then), and the presence of a neural network(s) in the supervisory MCU may improve reliability, safety and performance. For example, the diverse implementation and intentional non-identity makes the overall system more fault-tolerant, especially to faults caused by software (or software-hardware interface) functionality. For example, if there is a software bug or error in the software running on the primary computer, and the non-identical software code running on the secondary computer provides the same overall result, the supervisory MCU may have greater confidence that the overall result is correct, and the bug in software or hardware on primary computer is not causing material error.

(146) In some examples, the output of the ADAS system **538** may be fed into the primary computer's perception block and/or the primary computer's dynamic driving task block. For example, if the ADAS system **538** indicates a forward crash warning due to an object immediately

ahead, the perception block may use this information when identifying objects. In other examples, the secondary computer may have its own neural network which is trained and thus reduces the risk of false positives, as described herein.

(147) The vehicle **500** may further include the infotainment SoC **530** (e.g., an in-vehicle infotainment system (IVI)). Although illustrated and described as a SoC, the infotainment system may not be a SoC, and may include two or more discrete components. The infotainment SoC **530** may include a combination of hardware and software that may be used to provide audio (e.g., music, a personal digital assistant, navigational instructions, news, radio, etc.), video (e.g., TV, movies, streaming, etc.), phone (e.g., hands-free calling), network connectivity (e.g., LTE, Wi-Fi, etc.), and/or information services (e.g., navigation systems, rear-parking assistance, a radio data system, vehicle related information such as fuel level, total distance covered, brake fuel level, oil level, door open/close, air filter information, etc.) to the vehicle **500**. For example, the infotainment SoC **530** may radios, disk players, navigation systems, video players, USB and Bluetooth connectivity, carputers, in-car entertainment, Wi-Fi, steering wheel audio controls, hands free voice control, a heads-up display (HUD), an HMI display **534**, a telematics device, a control panel (e.g., for controlling and/or interacting with various components, features, and/or systems), and/or other components. The infotainment SoC **530** may further be used to provide information (e.g., visual and/or audible) to a user(s) of the vehicle, such as information from the ADAS system **538**, autonomous driving information such as planned vehicle maneuvers, trajectories, surrounding environment information (e.g., intersection information, vehicle information, road information, etc.), and/or other information.

(148) The infotainment SoC **530** may include GPU functionality. The infotainment SoC **530** may communicate over the bus **502** (e.g., CAN bus, Ethernet, etc.) with other devices, systems, and/or components of the vehicle **500**. In some examples, the infotainment SoC **530** may be coupled to a supervisory MCU such that the GPU of the infotainment system may perform some self-driving functions in the event that the primary controller(s) **536** (e.g., the primary and/or backup computers of the vehicle **500**) fail. In such an example, the infotainment SoC **530** may put the vehicle **500** into a chauffeur to safe stop mode, as described herein.

(149) The vehicle **500** may further include an instrument cluster **532** (e.g., a digital dash, an electronic instrument cluster, a digital instrument panel, etc.). The instrument cluster **532** may include a controller and/or supercomputer (e.g., a discrete controller or supercomputer). The instrument cluster **532** may include a set of instrumentation such as a speedometer, fuel level, oil pressure, tachometer, odometer, turn indicators, gearshift position indicator, seat belt warning light(s), parking-brake warning light(s), engine-malfunction light(s), airbag (SRS) system information, lighting controls, safety system controls, navigation information, etc. In some examples, information may be displayed and/or shared among the infotainment SoC **530** and the instrument cluster **532**. In other words, the instrument cluster **532** may be included as part of the infotainment SoC **530**, or vice versa.

(150) FIG. **5**D is a system diagram for communication between cloud-based server(s) and the example autonomous vehicle **500** of FIG. **5**A, in accordance with some embodiments of the present disclosure. The system **576** may include server(s) **578**, network(s) **590**, and vehicles, including the vehicle **500**. The server(s) **578** may include a plurality of GPUs **584**(A)-**584**(H) (collectively referred to herein as GPUs **584**), PCIe switches **582**(A)-**582**(H) (collectively referred to herein as PCIe switches **582**), and/or CPUs **580**(A)-**580**(B) (collectively referred to herein as CPUs **580**). The GPUs **584**, the CPUs **580**, and the PCIe switches may be interconnected with high-speed interconnects such as, for example and without limitation, NVLink interfaces **588** developed by NVIDIA and/or PCIe connections **586**. In some examples, the GPUs **584** are connected via NVLink and/or NVSwitch SoC and the GPUs **584** and the PCIe switches **582** are connected via PCIe interconnects. Although eight GPUs **584**, two CPUs **580**, and two PCIe switches are illustrated, this is not intended to be limiting. Depending on the embodiment, each of the server(s)

578 may include any number of GPUs **584**, CPUs **580**, and/or PCIe switches. For example, the server(s) **578** may each include eight, sixteen, thirty-two, and/or more GPUs **584**.
(151) The server(s) **578** may receive, over the network(s) **590** and from the vehicles, image data representative of images showing unexpected or changed road conditions, such as recently commenced road-work. The server(s) **578** may transmit, over the network(s) **590** and to the vehicles, neural networks **592**, updated neural networks **592**, and/or map information **594**, including information regarding traffic and road conditions. The updates to the map information **594** may include updates for the HD map **522**, such as information regarding construction sites, potholes, detours, flooding, and/or other obstructions. In some examples, the neural networks **592**, the updated neural networks **592**, and/or the map information **594** may have resulted from new training and/or experiences represented in data received from any number of vehicles in the environment, and/or based on training performed at a datacenter (e.g., using the server(s) **578** and/or other servers).
(152) The server(s) **578** may be used to train machine learning models (e.g., neural networks) based on training data. The training data may be generated by the vehicles, and/or may be generated in a simulation (e.g., using a game engine). In some examples, the training data is tagged (e.g., where the neural network benefits from supervised learning) and/or undergoes other pre-processing, while in other examples the training data is not tagged and/or pre-processed (e.g., where the neural network does not require supervised learning). Training may be executed according to any one or more classes of machine learning techniques, including, without limitation, classes such as: supervised training, semi-supervised training, unsupervised training, self learning, reinforcement learning, federated learning, transfer learning, feature learning (including principal component and cluster analyses), multi-linear subspace learning, manifold learning, representation learning (including spare dictionary learning), rule-based machine learning, anomaly detection, and any variants or combinations therefor. Once the machine learning models are trained, the machine learning models may be used by the vehicles (e.g., transmitted to the vehicles over the network(s) **590**, and/or the machine learning models may be used by the server(s) **578** to remotely monitor the vehicles.
(153) In some examples, the server(s) **578** may receive data from the vehicles and apply the data to up-to-date real-time neural networks for real-time intelligent inferencing. The server(s) **578** may include deep-learning supercomputers and/or dedicated AI computers powered by GPU(s) **584**, such as a DGX and DGX Station machines developed by NVIDIA. However, in some examples, the server(s) **578** may include deep learning infrastructure that use only CPU-powered datacenters.
(154) The deep-learning infrastructure of the server(s) **578** may be capable of fast, real-time inferencing, and may use that capability to evaluate and verify the health of the processors, software, and/or associated hardware in the vehicle **500**. For example, the deep-learning infrastructure may receive periodic updates from the vehicle **500**, such as a sequence of images and/or objects that the vehicle **500** has located in that sequence of images (e.g., via computer vision and/or other machine learning object classification techniques). The deep-learning infrastructure may run its own neural network to identify the objects and compare them with the objects identified by the vehicle **500** and, if the results do not match and the infrastructure concludes that the AI in the vehicle **500** is malfunctioning, the server(s) **578** may transmit a signal to the vehicle **500** instructing a fail-safe computer of the vehicle **500** to assume control, notify the passengers, and complete a safe parking maneuver.
(155) For inferencing, the server(s) **578** may include the GPU(s) **584** and one or more programmable inference accelerators (e.g., NVIDIA's TensorRT). The combination of GPU-powered servers and inference acceleration may make real-time responsiveness possible. In other examples, such as where performance is less critical, servers powered by CPUs, FPGAs, and other processors may be used for inferencing.
(156) Example Computing Device

(157) FIG. **6** is a block diagram of an example computing device **600** suitable for use in implementing some embodiments of the present disclosure. Computing device **600** may include a bus **602** that directly or indirectly couples the following devices: memory **604**, one or more central processing units (CPUs) **606**, one or more graphics processing units (GPUs) **608**, a communication interface **610**, input/output (I/O) ports **612**, input/output components **614**, a power supply **616**, and one or more presentation components **618** (e.g., display(s)).
(158) Although the various blocks of FIG. **6** are shown as connected via the bus **602** with lines, this is not intended to be limiting and is for clarity only. For example, in some embodiments, a presentation component **618**, such as a display device, may be considered an I/O component **614** (e.g., if the display is a touch screen). As another example, the CPUs **606** and/or GPUs **608** may include memory (e.g., the memory **604** may be representative of a storage device in addition to the memory of the GPUs **608**, the CPUs **606**, and/or other components). In other words, the computing device of FIG. **6** is merely illustrative. Distinction is not made between such categories as “workstation,” “server,” “laptop,” “desktop,” “tablet,” “client device,” “mobile device,” “handheld device,” “game console,” “electronic control unit (ECU),” “virtual reality system,” and/or other device or system types, as all are contemplated within the scope of the computing device of FIG. **6**.
(159) The bus **602** may represent one or more busses, such as an address bus, a data bus, a control bus, or a combination thereof. The bus **602** may include one or more bus types, such as an industry standard architecture (ISA) bus, an extended industry standard architecture (EISA) bus, a video electronics standards association (VESA) bus, a peripheral component interconnect (PCI) bus, a peripheral component interconnect express (PCIe) bus, and/or another type of bus.
(160) The memory **604** may include any of a variety of computer-readable media. The computer-readable media may be any available media that may be accessed by the computing device **600**. The computer-readable media may include both volatile and nonvolatile media, and removable and non-removable media. By way of example, and not limitation, the computer-readable media may comprise computer-storage media and communication media.
(161) The computer-storage media may include both volatile and nonvolatile media and/or removable and non-removable media implemented in any method or technology for storage of information such as computer-readable instructions, data structures, program modules, and/or other data types. For example, the memory **604** may store computer-readable instructions (e.g., that represent a program(s) and/or a program element(s), such as an operating system. Computer-storage media may include, but is not limited to, RAM, ROM, EEPROM, flash memory or other memory technology, CD-ROM, digital versatile disks (DVD) or other optical disk storage, magnetic cassettes, magnetic tape, magnetic disk storage or other magnetic storage devices, or any other medium which may be used to store the desired information and which may be accessed by computing device **600**. As used herein, computer storage media does not comprise signals per se.
(162) The communication media may embody computer-readable instructions, data structures, program modules, and/or other data types in a modulated data signal such as a carrier wave or other transport mechanism and includes any information delivery media. The term “modulated data signal” may refer to a signal that has one or more of its characteristics set or changed in such a manner as to encode information in the signal. By way of example, and not limitation, the communication media may include wired media such as a wired network or direct-wired connection, and wireless media such as acoustic, RF, infrared and other wireless media. Combinations of any of the above should also be included within the scope of computer-readable media.
(163) The CPU(s) **606** may be configured to execute the computer-readable instructions to control one or more components of the computing device **600** to perform one or more of the methods and/or processes described herein. The CPU(s) **606** may each include one or more cores (e.g., one, two, four, eight, twenty-eight, seventy-two, etc.) that are capable of handling a multitude of software threads simultaneously. The CPU(s) **606** may include any type of processor, and may

include different types of processors depending on the type of computing device **600** implemented (e.g., processors with fewer cores for mobile devices and processors with more cores for servers). For example, depending on the type of computing device **600**, the processor may be an ARM processor implemented using Reduced Instruction Set Computing (RISC) or an x86 processor implemented using Complex Instruction Set Computing (CISC). The computing device **600** may include one or more CPUs **606** in addition to one or more microprocessors or supplementary co-processors, such as math co-processors.

(164) The GPU(s) **608** may be used by the computing device **600** to render graphics (e.g., 3D graphics). The GPU(s) **608** may include hundreds or thousands of cores that are capable of handling hundreds or thousands of software threads simultaneously. The GPU(s) **608** may generate pixel data for output images in response to rendering commands (e.g., rendering commands from the CPU(s) **606** received via a host interface). The GPU(s) **608** may include graphics memory, such as display memory, for storing pixel data. The display memory may be included as part of the memory **604**. The GPU(s) **608** may include two or more GPUs operating in parallel (e.g., via a link). When combined together, each GPU **608** may generate pixel data for different portions of an output image or for different output images (e.g., a first GPU for a first image and a second GPU for a second image). Each GPU may include its own memory, or may share memory with other GPUs.

(165) In examples where the computing device **600** does not include the GPU(s) **608**, the CPU(s) **606** may be used to render graphics.

(166) The communication interface **610** may include one or more receivers, transmitters, and/or transceivers that enable the computing device **600** to communicate with other computing devices via an electronic communication network, included wired and/or wireless communications. The communication interface **610** may include components and functionality to enable communication over any of a number of different networks, such as wireless networks (e.g., Wi-Fi, Z-Wave, Bluetooth, Bluetooth LE, ZigBee, etc.), wired networks (e.g., communicating over Ethernet), low-power wide-area networks (e.g., LoRaWAN, SigFox, etc.), and/or the Internet.

(167) The I/O ports **612** may enable the computing device **600** to be logically coupled to other devices including the I/O components **614**, the presentation component(s) **618**, and/or other components, some of which may be built in to (e.g., integrated in) the computing device **600**. Illustrative I/O components **614** include a microphone, mouse, keyboard, joystick, game pad, game controller, satellite dish, scanner, printer, wireless device, etc. The I/O components **614** may provide a natural user interface (NUI) that processes air gestures, voice, or other physiological inputs generated by a user. In some instances, inputs may be transmitted to an appropriate network element for further processing. An NUI may implement any combination of speech recognition, stylus recognition, facial recognition, biometric recognition, gesture recognition both on screen and adjacent to the screen, air gestures, head and eye tracking, and touch recognition (as described in more detail below) associated with a display of the computing device **600**. The computing device **600** may be include depth cameras, such as stereoscopic camera systems, infrared camera systems, RGB camera systems, touchscreen technology, and combinations of these, for gesture detection and recognition. Additionally, the computing device **600** may include accelerometers or gyroscopes (e.g., as part of an inertia measurement unit (IMU)) that enable detection of motion. In some examples, the output of the accelerometers or gyroscopes may be used by the computing device **600** to render immersive augmented reality or virtual reality.

(168) The power supply **616** may include a hard-wired power supply, a battery power supply, or a combination thereof. The power supply **616** may provide power to the computing device **600** to enable the components of the computing device **600** to operate.

(169) The presentation component(s) **618** may include a display (e.g., a monitor, a touch screen, a television screen, a heads-up-display (HUD), other display types, or a combination thereof), speakers, and/or other presentation components. The presentation component(s) **618** may receive

data from other components (e.g., the GPU(s) **608**, the CPU(s) **606**, etc.), and output the data (e.g., as an image, video, sound, etc.).
(170) The disclosure may be described in the general context of computer code or machine-useable instructions, including computer-executable instructions such as program modules, being executed by a computer or other machine, such as a personal data assistant or other handheld device. Generally, program modules including routines, programs, objects, components, data structures, etc., refer to code that perform particular tasks or implement particular abstract data types. The disclosure may be practiced in a variety of system configurations, including handheld devices, consumer electronics, general-purpose computers, more specialty computing devices, etc. The disclosure may also be practiced in distributed computing environments where tasks are performed by remote-processing devices that are linked through a communications network.
(171) As used herein, a recitation of “and/or” with respect to two or more elements should be interpreted to mean only one element, or a combination of elements. For example, “element A, element B, and/or element C” may include only element A, only element B, only element C, element A and element B, element A and element C, element B and element C, or elements A, B, and C. In addition, “at least one of element A or element B” may include at least one of element A, at least one of element B, or at least one of element A and at least one of element B. Further, “at least one of element A and element B” may include at least one of element A, at least one of element B, or at least one of element A and at least one of element B.
(172) The subject matter of the present disclosure is described with specificity herein to meet statutory requirements. However, the description itself is not intended to limit the scope of this disclosure. Rather, the inventors have contemplated that the claimed subject matter might also be embodied in other ways, to include different steps or combinations of steps similar to the ones described in this document, in conjunction with other present or future technologies. Moreover, although the terms “step” and/or “block” may be used herein to connote different elements of methods employed, the terms should not be interpreted as implying any particular order among or between various steps herein disclosed unless and except when the order of individual steps is explicitly described.

Claims

1. A method comprising: obtaining map data representing a map, the map including one or more first features at one or more first locations within an environment and one or more first labels that classify one or more feature types associated with the one or more first features; obtaining image data representing an image and generated using one or more sensors of a dynamic actor; determining, based at least on a localization of the dynamic actor at a time with respect to the map, that one or more second features depicted by the image include the one or more first features represented by the map; based at least on the determining that the one or more second features depicted by the image include the one or more first features represented by the map, generating ground truth data by at least: correlating the one or more first locations of the one or more first features from the map with one or more second locations within the image that are associated with the one or more second features; generating, based at least on the correlating, one or more bounding shapes or one or more line segments for the image that are indicative of the one or more second locations within the image; generating, using the one or more first labels from the map, one or more second labels for the image that classify the one or more feature types associated with the one or more second features; and generating the ground truth data to represent the one or more bounding shapes or the one or more line segments and the one or more second labels; and updating one or more parameters of one or more neural networks based at least on a comparison between the ground truth data and one or more predictions of the one or more neural networks corresponding to the image.

2. The method of claim 1, further comprising: determining, using the one or more neural networks and based at least on the image data, one or more predicted bounding shapes indicating one or more predicted locations of the one or more second features depicted by the image, wherein the updating the one or more parameters of the one or more neural networks is based at least on a comparison of the one or more predicted bounding shapes and the one or more bounding shapes represented by the ground truth data.
3. The method of claim 1, further comprising localizing the dynamic actor based at least on at least one of global navigation satellite system (GNSS) data, global positioning system (GPS) data, or differential GPS (DGPS) data generated using one or more location-based sensors of the dynamic actor.
4. The method of claim 1, wherein the updating the one or more parameters of the one or more neural networks comprises updating the one or more parameters of the one or more neural networks such that the one or more neural networks perform inferencing for at least one of: object detection, feature detection, road feature detection, wait condition detection or classification, or future trajectory generation.
5. The method of claim 1, further comprising: receiving data representative of a trajectory of the dynamic actor through at least a portion of a field of view corresponding to the image data; and adjusting the trajectory based at least on the map.
6. The method of claim 5, wherein the adjusting the trajectory includes shifting the trajectory toward a rail of a lane of travel of the dynamic actor.
7. The method of claim 1, wherein: the one or more second locations comprise one or more pixels of the image; and the one or more bounding shapes or one or more line segments indicate the one or more pixels of the image.
8. The method of claim 1, further comprising: transforming the map from a first coordinate system of the map to a second coordinate system of the image data, wherein the ground truth data corresponds to the second coordinate system.
9. The method of claim 8, wherein the first coordinate system is a three-dimensional (3D) world-space coordinate system and the second coordinate system is a two-dimensional (2D) image-space coordinate system.
10. The method of claim 1, further comprising: determining an orientation of the dynamic actor with respect to the map, wherein the determining the one or more second features depicted by the image that include the one or more first features represented by the map is further based at least on the orientation.
11. The method of claim 1, wherein the updating the one or more parameters of the one or more neural networks comprises: determining, using one or more loss functions, one or more differences between one or more predicted locations determined using the one or more neural networks and the one or more second locations; and updating the one or more parameters based at least on the one or more differences.
12. The method of claim 1, further comprising: determining one or more vertices associated with the one or more bounding shapes indicative of the one or more second locations within the image that depict the one or more second features, wherein the generating of the ground truth data uses the one or more vertices.
13. A method comprising: determining, based at least on a localization of a dynamic actor with respect to a map, that one or more first features represented by the map include one or more second features represented by an image; based at least on the determining that the one or more first features represented by the map include the one or more second features represented by the image, generating ground truth data by: correlating one or more first locations of the one or more first features within the map with one or more second locations of the one or more second features within the image; generating, based at least on the one or more first locations corresponding to the one or more second locations, one or more indications associated with the one or more second

locations within the image; generating one or more first labels, determined using one or more second labels for the one or more first features represented by the map, that classify the one or more second features in the image; and generating the ground truth data to represent the one or more indications of the one or more second locations and the one or more first labels; and updating one or more parameters of one or more neural networks based at least on the ground truth data.
14. The method of claim 13, further comprising: correlating, based at least on the localization, the map with the image, the correlating including transforming the map to a coordinate space oriented with respect to the dynamic actor, wherein the determining that the one or more first features represented by the map include the one or more second features represented by the image is based at least on the correlating.
15. The method of claim 14, wherein the correlating further includes, after the transforming, converting the map from world-space coordinates to sensor-space coordinates corresponding to the image.
16. The method of claim 13, wherein the updating the one or more parameters of the one or more neural networks is, at least, by: determining, using the one or more neural networks and based at least on image data representative of the image, one or more predicted pixels-locations within the image corresponding to the one or more second features and one or more predicted labels corresponding to the one or more second features; determining one or more first differences between the one or more predicted locations and the one or more second locations and one or more second differences between the one or more predicted labels and the one or more first labels; and updating the one or more parameters based at least on the one or more first differences and the one or more second differences.
17. A system comprising: one or more processing units to: establish a localization of a dynamic actor with respect to a map; obtain, using one or more sensors of the dynamic actor, image data representative of an image generated using the one or more sensors during the localization; based at least on the localization, correlate one or more first locations of one or more features represented by the map with one or more second locations of the one or more features in the image, the one or more features including at least one of static objects, road markings, traffic signals, or traffic signs; based at least on the correlation, generate one or more ground truth labels corresponding to the one or more second locations of the one or more features in the image; and update one or more parameters of one or more neural networks based at least on comparing the one or more ground truth labels with one or more predictions of the one or more neural networks corresponding to the image.
18. The system of claim 17, wherein the one or more processing units are further to generate one or more bounding shapes associated with the one or more second locations.
19. The system of claim 17, wherein the one or more parameters of the one or more neural networks are updated, at least, by: determining, using the one or more neural networks and based at least on the image data, the one or more predictions; determining one or more differences between the one or more predictions and the one or more ground truth labels; and updating the one or more parameters based at least on the one or more differences.
20. The system of claim 17, wherein the one or more second locations include one or more pixel locations within the image; and the one or more ground truth labels correspond to the one or more pixel locations.